

Chapter 1
Introduction

CHAPTER 1

INTRODUCTION

This study examines how corporate life stages significantly affect the capital structure of a firm at various stages of its life cycle. The three major decisions in corporate finance are investment, financing, and dividend decisions. Moreover, there are also working capital management decisions to consider. The study focuses on long-term financing, which is challenging to change quickly, as these decisions tend to be sticky.

Profitability and shareholder value have been the primary objectives of firms, guiding financing decisions, with shareholder value taking precedence over profitability in an ideal, efficient financial market scenario for many decades now. Furthermore, according to Aglietta (2000), shareholder value has become the prevailing standard for transforming capitalism, which has justified the widespread adoption of new policies and practices that favour shareholder value over other objectives of the firm.¹ Whether shareholders' value or other objectives of the firm, such as profit maximisation or something else, financing has been a major contributor to the fulfilment of those objectives in a substantial number of cases.

Finance has been a subject of inquiry in business ever since the concept of business firms emerged. Financing decisions, along with investment, dividend, and working capital management decisions, became more relevant with the emergence of corporate firms.

1.1. Capital structure

Capital structure, or the degree of financial leverage, is represented by the ratio of debt to equity or the ratio of debt to debt plus equity.

Capital structure refers to a firm's financing of its overall operations and growth through various sources of funds, primarily debt and equity. It is a critical component of a firm's financial strategy because it directly affects the risk and value of the business. A well-balanced capital structure can minimise the cost of capital, improve

¹ Aglietta, M (2000), Shareholders Value and Corporate Governance: Some Tricky Questions, *Economy and Society*, Vol. 29 (1), pp. 146-159.

profitability, and enhance shareholder value. The two main components are equity capital, which includes common stock, preferred stock, and retained earnings, and debt capital, which comprises short-term borrowings, long-term loans, bonds, and other forms of debt. An optimal capital structure maintains a proper mix of debt and equity to balance risk and return. High debt levels can lead to increased financial risk due to interest obligations and potential insolvency, especially during downturns. Still, they can also provide tax advantages because interest is tax-deductible. On the other hand, relying heavily on equity may dilute ownership and reduce earnings per share; however, it offers greater financial stability since dividends are not mandatory. The choice of capital structure is influenced by various factors, including the company's size, industry characteristics, cash flow stability, financing costs, control considerations, tax implications, market conditions, and management's attitude toward risk. Theories such as the Modigliani-Miller theorem, trade-off theory, pecking order theory, and agency theory provide different perspectives on how firms should approach capital structure decisions. Ultimately, a company's capital structure needs to be tailored to its specific circumstances and strategic goals, ensuring it can sustain operations, finance new projects, and maximise long-term value.

1.2. Implications of capital structure

In the corporate world, some firms are pure equity firms and remain so. Others, from their initial stage, utilise debt as a source of finance. Additionally, another kind moves from pure equity to those that use debt as a source of finance. Furthermore, it is possible that a fourth kind may exist, which, having initially used debt as a source of finance, could again become a pure equity firm. So, it becomes pertinent to understand how their capital structure decisions are guided or determined.

The implications of capital structure are far-reaching because how a company finances itself impacts its overall risk, cost of capital, and even its ability to grow. One of the most significant implications is how it affects the company's 'financial risk'. Suppose a company uses a lot of debt (borrowed money). In that case, it increases its risk because it must pay interest, whether it earns a profit or not, and eventually repay the loan, regardless of the business's performance. This is called 'financial leverage'.

When times are good, debt can boost profits because the company can invest the borrowed money and potentially earn more than the cost of borrowing. However, too much debt can make it harder to stay afloat during difficult times.

On the other hand, using ‘equity financing’ (issuing shares or utilising retained earnings) tends to reduce risk because there are no mandatory payments, unlike with debt financing. However, equity comes with its challenges, such as diluting ownership and reducing control for existing shareholders. If a company issues too much equity, it may also lower the value of each share, which could affect shareholders' long-term interests.

Another important implication is the ‘cost of capital’. Companies must pay for the capital they use, whether it is interest on debt, efforts to generate appreciation in the value of capital, or dividends on equity. Generally, debt tends to be cheaper than equity because it is less risky for the investor (interest payments are predictable, and debt holders get paid before equity holders). However, the more debt a company has, the higher the interest rate it may pay, as lenders perceive it as riskier. This means that companies must carefully balance their debt and equity levels to minimise their overall cost of capital. Therefore, there is a trade-off between funding sources from equity and debt.

The capital structure also impacts a company’s ability to grow. A company with a strong balance between debt and equity might expand more easily by securing loans at lower interest rates while maintaining control. However, if a company takes on too much debt, it might be stuck, unable to borrow more or invest in new projects due to high interest payments. Conversely, a company that relies too heavily on equity may limit its growth potential by issuing too many shares, which could dilute the value of its stock.

Finally, the ‘market’s perception’ of a company’s capital structure is key. Investors often consider a company's financing when deciding whether to invest in it. If a company has a high level of debt, it may be perceived as riskier, even if it is managing its finances well. Conversely, suppose a company relies too heavily on equity. In that case, it may be perceived as lacking confidence in its business model and may be hesitant to take on debt, which could negatively impact stock prices.

In short, the implications of capital structure extend beyond mere numbers. It affects the company's risk profile, its ability to invest and grow, its cost of financing, and how investors view it. Finding the right balance is crucial because it can significantly impact a business's long-term success or failure.

1.3. Firm's stages of life

From the conception of an idea to the ultimate decline of a firm, it may undergo many stages in its life cycle, including startup, growth, maturity, and decline. Depending upon the nature of the product or industry, the timeline may vary.

Greiner's Growth Model describes the phases that organisations go through as they grow. All types of organisations, from design shops to manufacturers, construction companies to professional service firms, experience these phases. Each growth phase consists of a period of relatively stable growth, followed by a "crisis" when significant organisational change is necessary for the company to continue growing.

Larry E. Greiner originally proposed this model in 1972, which consists of five stages of growth. In 1998, he added a sixth stage in an updated version of his original article. The six growth stages are described below: growth through creativity, direction, delegation, coordination and monitoring, collaboration, and extra-organisational solutions.

Anthony and Ramesh's (1992) and Dickinson's (2011) research have been significant milestones in classifying a firm's life stages. We will discuss these developments in classifying life stages in later chapters.

1.4. Motivation for the study

As part of my research coursework, I had the opportunity to study Altman's (1968) work on insolvency and bankruptcy. This endeavour prompted me to examine the evolution of some firms over their lifetimes, or, if we may use that term in the context of an artificial entity. Over time, while reviewing the literature, we found that the

determinants of capital structure across corporate life stages have been a relatively less-investigated area and may be worth further investigation.

Not only are financial needs different at each stage of a firm's life, but the availability of financing sources also varies. Not only is it a constraint for a business firm to conform to the context, but it is also a policy question to enhance the financial system along the desired lines. For example, the growth of startups may be significant for accelerating economic growth and development. Likewise, preserving enterprise capital is equally important in how we deal with declining firms and in strategising financing for business revival. During growth and maturity, profits are visibly substantial, and therefore, the financial system may be functioning well; however, an additional policy push may lead to accelerated growth.

1.5. Research objectives and research questions

Research objectives of the study are as follows:

To examine whether capital structure varies at different stages of a business firm's corporate life and determine whether distinct determinants of capital structure exist at various stages.

Research questions

Research questions arising from the research objectives are: How does capital structure differ at different life stages? What are the determinants of capital structure across different stages of corporate life? Do these determinants differ across different stages? Additionally, it will identify the determinants at start-up, growth, maturity, and decline stages and investigate whether the 'global financial crisis' has impacted capital structure. Furthermore, it seeks to examine whether a change in the bankruptcy regime brought a change in capital structure.

We will delve into it in more detail in our chapter on research design.

1.6. Scope of the study

Business firms go through various stages during their life cycles. The study aimed to investigate how capital structure and its determinants differ across the multiple stages of the corporate life cycle: startup, growth, maturity, and decline.

The financial data of Indian firms, excluding non-financial firms, have been used in the study. Therefore, the insights from the study may be more useful for the Indian corporate sector, specifically for firms outside of banking and other financial sectors.

1.7. Structure of the study

A brief outline of the chapters of the thesis is presented below:

The next chapter focuses on the theoretical perspective of the research subject. Chapter three presents a review of the literature to define the research focus of the present study more precisely. The research design, focusing on data and research methodology, is discussed in Chapter 4. Chapters five to eight highlight the main results of the study. The final chapter presents recommendations and limitations, followed by a summary of the thesis. The study concludes with a bibliography of articles and books in the annexure.

The study comprises nine chapters, each covering a distinct aspect of the research. A summary description of these chapters is provided below.

Chapter 2 covers the various capital structure theories that provide explanations for the heterogeneous behaviour of firms in maintaining the leverage ratio or capital structure and sets a theoretical framework for further empirical research in the area of capital structure. In this chapter we have discussed the irrelevance of capital structure theory by Modigliani and Miller (1958), corporate taxes and the capital structure by Modigliani and Miller (1963), debt and taxes by Miller (1977), trade-off theory by Kraus and Litzenberger (1973), signalling theory by Spence (1973), agency cost theory by Jensen and Meckling (1976) and pecking order theory by Myers and Majluf (1984). These theories provide arguments and counterarguments, as well as a framework for assessing the value relevance of a firm's capital structure decisions. They also explain how the

context and content of these decisions change at different stages of corporate life, from the incubation of ideas to start-up, growth, maturity, decline, and decay.

In Chapter 3, we provide a review of the existing literature to gain a deeper understanding of the work that has already been done by researchers in the area of capital structure and its determinants across corporate life stages. This review aims to identify research gaps that require further investigation, thereby informing the development of our research hypothesis. In this chapter, we conducted an existing literature review, covering literature from both developed and developing economies, with a focus on studies that include sample firms from India and its neighbouring countries.

Chapter 4 discusses the research methodology framework within which this study is conducted. We have described the data collection method, details of the variables, and their measurements used in the study, followed by a list of research hypotheses developed for the study. Finally, we have discussed the data analysis techniques that we have applied to test our research hypotheses. Specifically, we have discussed pooled ordinary least squares (OLS) estimation, fixed effects estimation, random effects estimation, and dynamic panel regressions based on system generalised method of moments (GMM).

Chapter 5 presents the broad features of our sample and attempts at ANOVA analysis. We investigate whether capital structure differs at various stages of corporate life. We use both static and dynamic models. We examine both the determinants of capital structure and the factors influencing changes in capital structure.

Chapter 6 addresses the determinants of capital structure during the startup stage, which requires special treatment due to its distinct characteristics and financial requirements.

Chapter 7 examines the determinants of capital structure during both the growth stage and the maturity stage. As a firm gains market acceptance for its products, it enters the growth phase, and its financing needs increase disproportionately. The question of

which sources of finance should be used in preference to others becomes highly relevant. There are trade-offs to the choices that firms may make.

Chapter 8 deals with determinants of capital structure during the decline and decay stages. At some point in a business firm's life, it may encounter difficulties in surviving due to competition, obsolescence, or leadership succession.

Chapter 9 summarises the study, provides recommendations, discusses the study's limitations, and outlines the further scope of research.

Chapter 2
Theoretical Perspective

CHAPTER 2

THEORETICAL PERSPECTIVE

Having introduced the topic of the thesis in the previous chapter, we will present the theoretical perspective relating to determinants of capital structure or financial leverage in this chapter. We will interchangeably use these terms – capital structure and financial leverage. At the outset, we are concerned with long-term debt in this study, as it is a sticky and challenging aspect to change its quantum with the flip of a coin.

2.1. Introduction

Corporates are legal entities with a profit motive and a business mission. This mission is what corporates invest in, and the profit is what they reap. A share of this profit, as well as the increase in wealth they receive as a dividend and through the market valuation of their investments. However, corporates are not like a single species that is static and quickly becoming extinct, but rather a diverse group that evolves and is shaped by the changing economic and social structure of society.

We focus on corporates.² In India, from 2001 to 2024, our discussion will focus on the context of that period.

An entrepreneur floats a firm in the setting that is before him, or the setting that they choose to create. By setting, we mean the socio-economic framework or milieu³. The entire energy as an entrepreneur is directed to the goal that one sets for oneself. Still, the driving force for a business firm is the index of survival, the index of which is nothing but the residual that you deduct from your resources or revenue that is left after deducting the cost of short or long-term resources used in the endeavour. The firm that the entrepreneur floats derives its force from the entrepreneurial venture, the nature and scope of the industry, and the economic context in which it operates. These three forces have their dynamics. Our study is focused on the firm. Depending on how these forces interact, a firm assumes its shape over its lifetime. However, there is an

² To be used interchangeably with corporate firm, firm, or business firm.

³ Schumpeter, Joseph A. (1935), *The Theory of Economic Development: An Enquiry into Profit, Capital, Credit, Interest and the Business Cycle*. (Oxford University Press, 1935), p. 65.

internal/intrinsic force within the firm, which, although not isolated from the other or external forces, does have some autonomy in its behaviour. The firm continues to operate over decades and centuries, despite changes in the entrepreneur, industry, or economic setting, which may occur due to shifts in management or the economic environment. At times, the firm becomes diversified, or the regulatory regimes change due to political-legal shifts, or the industry technology within the firm is undergoing changes, or sometimes there is a change in entrepreneurial spirit, resulting from generational changes. So, the primary task is to identify characteristics of the firm which are intrinsic to it and evolve over its life, or one may even use the term life stages, if it is feasible to categorise the life of the firm into different stages, based on specific characteristics, like age, technology or cash flow patterns.

Since our focus is corporate finance, we will confine our exposition to that theme. Corporate finance has some well-defined contours around what we describe as finance decisions. That is, investment, financing, dividend, and working capital management decisions. The present study examines the factors that determine a firm's leverage, also known as its financial leverage or capital structure. Studies on the determinants of capital structure are numerous and varied, ranging from the relevance of the decision to valuation of the firm, to the trade-off, agency, pecking order, and dynamic relationships throughout corporate life stages. The next chapter will survey the literature to define the scope of this study's research more precisely.

MM (1958) focused on applying some of the standard tools of economics to corporate finance problems, particularly the analysis of competitive market equilibrium. Before that time, academic discussions in finance were primarily focused on the empirical issue of what the market capitalised. Did the market capitalise a firm's dividends, earnings, or some weighted combination of the two? Did it capitalise net earnings, net operating earnings, or something in between? The answers to these and related questions about the behaviour of interest rates were supposed to provide a basis for choosing an optimal capital structure for the firm in a framework analogous to the economist's model of discriminating monopsony.⁴

⁴ Miller Merton H (1977), Debt and Taxes, *The Journal of Finance*, Vol. 32 No. 2, May 1977, pp. 261-275.

2.2. Nature of economic and econometric relationship of the study

Our study typically involves a set of dependent variables and an independent variable. That is, the dependent variable Y is a function of explanatory variables X_i

$$Y = f(X)$$

Initially, let us consider the explanatory variable X_1 ; however, economic theory is uncertain about the other variables that may be explanatory, such as $X_2, \dots, X_k$. We start by assuming the simplest possible relationship in mathematical form.

$$Y = b_0 + b_1X_1$$

We compute b_0 and b_1 , referred to as estimates, using a sample.

Suppose we have 100 pairs of observations on X and Y , the scatter diagram of these observations never aligns exactly on a line, in a meaningful economic relationship. The deviations of the observations from the line may be attributed to several factors.

In economic reality, each variable is influenced by many factors. For instance, a family's consumption pattern is determined by several factors, including family income, prices, the composition of the family by age and sex, level of education, tastes, religion, social and educational status, wealth, and others. One could compile an almost non-ending list of such factors. However, not all the factors influencing a particular variable can be included in the function for various reasons.

Some of the factors may not be known even to the person most acquainted with the relationship being studied. This lack of knowledge is, to a greater extent, due to an incomplete understanding of the theory regarding the variation of economic variables in general.

Even when known to be relevant, some factors cannot be measured statistically. These are mainly psychological factors, or, in general, qualitative factors (such as tastes, expectations, and religion). For example, some companies have a history of not leveraging themselves and maintaining a zero-debt status. This could be due to the mindset of the promoters or top management. However, post-COVID-19, some of these have turned to debt markets for financing, due to difficulties these companies

might have faced due to a lack of availability of internal resources or finances, a lack of opportunity to tap capital markets directly for equity, or perhaps the government was willing to lend a helping hand in such times of global crisis.

Some factors are random, appearing in an unpredictable manner and at unpredictable times, making it difficult to account for their influence satisfactorily (e.g., epidemics, earthquakes, wars).

Some other factors may individually have a negligible influence on the dependent variable. Thus, their parameter is so small that it cannot be reliably measured due to rounding errors in the computations. However, all these factors together may account for a considerable part of the variation in the dependent variable.

Even if all factors are known, the available data are often insufficient to measure all factors that influence the relationship between a dependent variable and the independent variables. This is particularly so when we use time series, which are usually short.

Thus, in most cases, only a few of the most important variables are explicitly included in the function.

On an XY graph, the scatter of points around the line may be attributed to an erratic element inherent in human-caused deviations from the 'normal' behavioural pattern depicted by the line. For example, in a moment's whim, a consumer changes their expenditure pattern, although income and prices have not changed. To complete the argument, on this aspect, it could also be due to some factor not yet known, and not necessarily a moment's whim.

An imperfect specification of the mathematical form of the model could also result in estimation errors. We may have linearised the relationship between dependent and independent variables, while the relationship might not be linear.

The first four sources of error – $err1$, $err2$, $err3$, and $err4$ - render the form of the equation wrong, and they are usually referred to as errors in the equation or errors of

omission. The fifth source of error is called the error of measurement or observation. It is usual, of course, to have both these types of error simultaneously in the function.

To consider the above sources of error we introduce in econometric functions a random variable which is usually denoted by the letter u and is called error term or random disturbance or stochastic term of the function, so called because u is supposed to 'disturb' the exact linear relationship which is assumed to exist between X and Y . By introducing this random variable in the function, the model is rendered stochastic of the form.

$$Y = b_0 + b_i X_i + u$$

The genuine XY relationship, which connects the variables involved in a linear relationship, is divided into two parts: a part represented by a line and a part defined by the random term u , representing systematic variation and random variation, respectively. Alternatively, it can be represented by variation in the dependent variable, as defined by explained variation and unexplained variation, respectively.

2.3. The capital structure decisions and theories

To begin with, a list of factors may be drawn which affect financing decisions of firms: the goal of the firm, the 'typical' financial leverage of similar firms in the industry, availability of internal funds: the rate of growth of earnings, availability of internal funds: the retention policy of managers, degree of concentration of ownership and voting control and credit limits (or debt capacity) of the firm depending on the size of the firms, the growth of the assets (size) of the firm, stability of earnings, asset structure, the cost of debt, the cost of equity financing, the corporate tax rate, expectations regarding the rate of inflation, availability of loanable funds, and general customs in capital markets.

There are many theories which are based on the objective of stockholder-wealth maximisation, that deserve consideration: classical or traditional, developed systematically by Ezra Solomon, postulates that there is range of optimal capital structures, that is, a range of debt/equity ratios over which the discount rate (or the average cost of capital) attains its minimum value, resulting in the maximisation of

the market value of the firm (over that range). All these theories examine the question of whether there is a capital structure that maximises the value of the firm to the existing (current) stockholders.

The theory developed by Modigliani and Miller (MM), under the assumption of perfect capital markets and no corporate (or other) taxes, postulates that the value of the firm, and consequently, the wealth positions of stockholders, is not affected by the capital structure (type of financing).

As a modification, it considers corporate taxes and postulates that a firm should use as much debt as possible to maximise its value and the wealth position of its stockholders

2.3.1. Managerial theory

How managers in practice decide the proportion of debt and equity in the firm's capital structure.

A managerial theory of capital structure

The managerial school (Baumol, Galbraith, Marris, Williamson) posits that managers prioritise the maximisation of their utility rather than the maximisation of the utility of owner-shareholders. The basis of the model is the Marris-Galbraith model, which assumes a managerial utility function containing two variables: the job security of management as a whole and the firm's growth rate. Given ample evidence that large firms are growth seekers, the study focuses on the job security variable. In particular, the following job-security hypothesis or group-behaviour hypothesis is postulated:

The capital structure of firms is decided to safeguard the job security of managers.

Managers believe that their job security is safeguarded if the firm's debt-equity ratio does not deviate significantly from the typical industry debt-equity ratio in the long run.

The use of debt is influenced by managers' preference for internal finance, which is primarily defined by the growth of earnings and the firm's retention policy. However,

the job-security considerations set a limit to the amount of profits that managers can retain. Given that managers are growth-seekers, internal funds are inadequate to cover the financial needs of a fast rate of growth. Hence, managers resort to external sources of funds. They prefer debt over the issue of new stock or equity. The cost considerations of these two sources of funds, as well as the possibility of compromising voting control in a way that endangers managerial job security, make debt a more attractive source of external finance.

The creditors' attitudes temper the desires of managers regarding the use of debt. Creditors ultimately determine the debt capacity of the firm, considering various factors, such as the firm's size (in terms of assets), its future growth potential, and the variability of its earnings, among others. Furthermore, creditors determine the amount of funds that they are willing to offer at various (largely externally determined) interest rates, depending on their expectations about the rate of inflation.

From the above considerations, in a managerialist framework the determinants of the capital structure of firms would be: a) the typical 'debt/equity ratio'; b) the past level of capital structure of the firms; c) the size (assets) of the firm; d) the growth of assets of the firm; e) the stability of earnings; f) the expected rate of inflation; g) the rate of growth of earnings; h) the retention ratio; i) the cost of debt; j) the cost of new issue of stock; k) the corporate tax rate; and l) the dispersion of ownership of shares.

The study by Koutsoyiannis (1978) employed the empirical specification with the OLS model to analyse thirty cross-section samples from five US industries: textiles, paper products, electrical equipment and appliances, steel, and food. The study covered the period from 1960 to 1974. *A priori*,

the debt-equity ratio was positively associated with the growth rate of tangible capital assets and book value of real assets of the firm (+vely related)

Coefficients of lag dependent variable of the firm and the lag variable of the debt-equity ratio of similar firms in the industry were positive but less than one (between 0 and 1)

debt-equity ratio was negatively related with standard deviation of the total EBIT of the firm, shareholders' control, measured by the number of shares per stockholder, the rate of growth of earnings of the firm, the retention ratio of the firm, and percentage price change

of the share of the firm over the previous period, as a proxy for the expected cost of equity capital. (-very related)

The results of such efforts provide support for the ‘managerial job-security hypothesis’. There is partial adjustment of the actual debt/equity ratio of the firm to the ‘average’ leverage of firms in the industry, which may be subject to externally imposed credit limits by lenders, and to the ease of access of managers to internal funds (retained earnings).

In particular, the results suggest that:

At times, large firms set a target debt/equity ratio (D/E) based on the ‘typical’ or ‘average’ leverage of similar firms in the industry. The capital-structure decisions of managers are closely related to the leverage policy of their rival firms in the industry, because managers feel safe if they do not deviate in the long run from the norms established in the industry. Here, one can distinguish between the herding and innovative behaviour of industry captains and the risks associated with these tendencies. However, that was not the focus of Koutsoyiannis's study (1978).

One of the most important determinants of the debt-equity ratio is the firm’s ‘debt capacity’, which is essentially a debt limit set by lenders in the capital market and depends on the firm’s growth potential. However, in that sense, a borrowing firm’s financial risk is determined within the constraint set by lenders.

Often, the availability of internal funds, which can influence the rate of growth, the amount of earnings, and the dividend payout or overall payout policy of management, is more important than the issuance of new stock in determining the use of debt.

Contrary to common belief, neither the variability of earnings nor the size of the firm appears to be a significant explanatory variable for the capital structure of large corporations.

The share price variable is the least significant of all regressors. This result supports the view that management does not consider the issue of new stocks as an attractive alternative to debt, and it is consistent with the observed value of new stock issues relative to those of bonds during the study period. (1960-74).

As one can see, the ideas on trade-off theory (TOT) and pecking order theory (POT) began to germinate here. There may be aspects of adverse selection, agency conflict, information asymmetry, moral hazard, as well as planned behaviour, which could be explored here. Along with actors in decision-making, their self-interest gets reflected in the way decisions are made. Here, the focus is on management in firms.

The financial leverage decision in practice

In the real business world, two methods are widely used as sources of relevant information about the alternative sources of funds: EPS-EBIT analysis and cash flow analysis.

2.3.1.1. *EPS-EBIT analysis*

It assumes that investors are primarily interested in the size of the earnings per share (EPS), that is, the yield of their investments. However, investors' decisions are based on both the expected return and the risk of the securities.

EPS-EBIT analysis involves determining the level of earnings per share (EPS) at various levels of total earnings before interest and taxes (EBIT $\equiv$) under different methods of financing the firm's operations.

Earnings per share are to be calculated by deducting interest on debt from earnings before interest and tax (EBIT) and further deducting tax out of this residual and dividing this final residual by N, where N is equal to the number of paid-up shares, i.e.,

$$EPS = \frac{(EBIT - i * Debt)(1 - tc)}{N}$$

2.3.1.2. *Cash flow analysis*

It involves the estimation of total earnings before interest and taxes (EBIT). This, in turn, requires a forecast of the stream of cash receipts (inflows) and cash payments for the production of the output (cash outflows). The difference is the total earnings (EBIT) from which fixed charges must be met. Forecasts of this sort are called 'cash

budgets' and can extend over any number of periods. Predictions are usually made under a single set of assumptions concerning general (external) business conditions and particular circumstances expected to prevail in the markets in which the firms operate. However, a cash flow analysis should include a forecast of cash inflows and cash outflows under different sets of assumptions. For example, a firm might examine the EBIT under the following 'states of business': severe recession, mild recession, expected growth, very rapid growth, and inflation.

Such forecasts, combined with subjective probabilities for each 'state of business', give probability distributions of earnings for each period. It is these distributions that provide information about probable deviations in actual earnings from their expected values. Hence, all the managers should be prepared for possible downward deviations that may lead to insolvency.

Given the probability distributions of the stream of cash flows, management can determine the amount of fixed charges and debt that the firm can undertake without running a great danger of insolvency. In many cases, management defines a 'tolerance limit' for insolvency, that is, a maximum value for the probability of the firm's earnings falling below its fixed charges.⁵

Cash-flow analysis is most useful for internal purposes. It helps hedge management against the risk of insolvency by providing a warning of difficult times for the company. However, insolvency is only one aspect of the financial risk of shareholders. The other aspect of this risk, namely the danger of having nothing left over for dividends after the fixed charges are met, is not considered by cash-flow analysis. Investors are most interested in the latter risk, assuming the firm will always have enough earnings to cover its fixed charges.

We will return to cash flow analysis later in the Review of Literature and Research Design chapters. Cash flow analysis is significantly important when we revisit the

⁵ For a detailed discussion of cash-flow analysis, see G Donaldson, *Corporate Debt Capacity*, Harvard Business School, 1961. Also, G. Donaldson, "Strategy for Financial Emergencies," *Harvard Business Review*, 1969, pp. 67-79.

primary focus of our study, that is, the corporate life stages as a variable of interest that determines capital structure as well as changes in capital structure.

2.3.2. Traditional theory

The financial life stages of corporates encompass dimensions such as trade-offs, pecking order, managerial aspects, information asymmetry, signalling of information, and free cash flows.

It is generally not feasible to incorporate all the factors that have a bearing on decisions, in this case, financing decisions, or financial leverage decisions, or capital structure decisions, in the study that examines the phenomenon, and, therefore, we end up with various theories: MM, TOT, POT, CLS. Therefore, some studies propose a theory, and others replicate and extend these studies with new data, countries, industries, and other explanatory variables. Sometimes, these studies develop something that could be a new theory, like a trade-off or a pecking order. Interwoven with others, such as information asymmetry, corporate life stages, and trade-off/pecking order.

There could be pure-equity firms that are born that way and go through all the stages of their life in terms of cash flow patterns, and may also become extinct. We are yet to break into their DNA. A recent study⁶,

In the sections that follow, each of these theories is briefly described to shape the direction of research that this study intends to undertake:

Based on the goal of stockholder-wealth maximisation, three main theories deserve consideration. All three examine the question of whether there is a capital structure that maximises the value of the firm and that of the existing stockholders. The first theory, known as the 'classical' or 'traditional' theory, postulates that there is a unique optimal capital structure that maximises the firm's market value by minimising the average cost of its capital. A version of this theory, developed systematically by Ezra

⁶ Hartzmark, Samuel M, & Solomon, David H (2019). Dividend Disconnect, The Journal of Finance, October 2019, Vol 74 (4), 2153-2199

Solomon⁷, postulates that there is a range of optimal capital structures, that is, a range of debt/equity ratios over which the discount rate (or the average cost of capital) attains its minimum value, resulting in the maximisation of the market value of the firm (over that range). He states, “The proper uses of debt financing are one of the major decision areas of corporate financial management. My paper confines itself to just one facet of the many considerations which jointly determine the optimal use of debt-namely, the effect that a change in financial leverage has, or can be assumed to have, on a company’s cost of capital.”⁸

“To isolate the effect of leverage alone from the many other factors that may be involved in using debt wisely, it is useful to analyse in terms of a simplified model: Let X be a company that holds or acquires only one kind of asset. Each dollar invested in these assets generates a flow of operating earnings, before taxes, which provides a rate of return of k per annum of a given quality concerning the certainty of uncertainty with which it can be expected to occur. We will assume that this company may use any mixture of only two kinds of financing - pure, externally derived equity, on the one hand, and pure debt, on the other. Third, we assume that the structure of market capitalisation rates is given and that this structure does not change over time.”⁹

For the all-equity case, we have $k_e = k_o = k$. When debt is used, we have $k_e > k_o$ if all the residual profits are distributed among equity holders. Specifically, regardless of the valuation theory one embraces, the relationship $k_e = k_o + (k_o - k_i) B/S$ and $k_o = k_e S + k B$ must hold in an assumed situation where all the profits are among capital providers. The heart of the leverage question can now be stated as follows: What happens to V and k_o as we increase the degree of leverage (other things remaining unchanged) from Leverage zero to leverage 100 per cent?

There are two ways to alter leverage in the model: by changing the liability structure while keeping the asset structure constant, or by changing both the liability and asset

⁷ Ezra Solomon, *The Theory of Financial Management* (Columbia University Press, 1963). And Ezra Solomon, Leverage and the cost of capital, *Journal of Finance*, May 1963, Vol. 18 (2), pp. 273-79.

⁸ Ezra Solomon, Leverage and the cost of capital, *Journal of Finance*, May 1963, Vol. 18 (2), pp. 273-79, p.273.

⁹ Ezra Solomon, Leverage and the cost of capital, *Journal of Finance*, May 1963, Vol. 18 (2), pp. 273-79, p.273.

structures. However, to avoid contaminating the leverage effect, it is necessary to assume that each new asset acquired generates operating earnings, before tax, of the same size and quality as those produced by existing assets.¹⁰

2.3.3. Analysing Modigliani and Miller (MM)¹¹ approach

Ignoring the tax effect, the Modigliani-Miller position is that Company X's overall cost of capital, k , remains constant across all leverage levels, from zero debt to nearly 100% debt. If k_o^* represents the overall cost of capital for a more levered company and k_o represents the overall cost of capital for a less levered company in the same risk class, their basic thesis is that, except for the tax effect, k_o^* is constant for all levels of leverage.

It is helpful to divide their basic thesis into two component statements. Still ignoring the tax effect, these are (1) an increase in borrowing (and hence in leverage), no matter how moderate or 'judicious,' can never lower a company's cost of capital; (2) an increase in borrowing (and hence in leverage), no matter how immoderate or 'excessive' can never raise a company's cost of capital.

Almost all the analytical controversy generated thus far by Modigliani-Miller's thesis seems to have centred on the first of these two component statements and the proof put forward by its proponents. The proof offered by Modigliani-Miller is that a process akin to arbitrage, in which individual investors engage in 'home-made' leverage as a substitute for corporate leverage, will keep k_o^* , the capitalisation rate for the more levered situation, equal to k_o , the capitalisation rate for the less levered situation.

Modigliani and Miller developed the second theory.¹² Under the assumptions of perfect capital markets and no corporate (or other) taxes, this theory postulates that the firm's value and, consequently, the wealth position of stockholders, is not affected by the capital structure (type of financing).

¹⁰ Ezra Solomon (1963), Leverage and the cost of capital, **Journal of Finance**, May 1963, Vol. 18 (2), pp. 273-79, p.275.

¹¹ Ezra Solomon (1963), Op. cit., pp. 273-279, p. 275

¹² F. Modigliani and M. Miller (1958), 'The Cost of Capital, Corporate Finance and the Theory of Investment', **American Economic Review**, 1958, pp. 261-77.

The third theory was also developed by Modigliani and Miller.¹³ It considers corporate taxes and postulates that a firm should use as much debt as possible to maximise its value and the wealth position of its stockholders.

The capital structure of firms is determined under the assumption that the goal of managers is to maximise stockholder wealth.

2.3.4. Pecking order theory (POT) and conditional pecking order

Since the seminal MM (1958) paper, much work has focused on testing the implications of the two traditional views of capital structure: the trade-off model and the pecking order model. In the trade-off model, firms presumably form a leverage target that optimally balances various costs and benefits. In the pecking order model, firms follow a financing hierarchy designed to minimise adverse selection costs of security issuance. Each theory succeeded in explaining general observed capital structures, such as the association between leverage and various firm characteristics, as well as the use of different sources of capital. However, neither has managed to explain much of the observed heterogeneity in capital structures, leverage changes, or other phenomena, including why leverage levels are so sticky.

Two competing theories dominate the discourse among researchers: the trade-off theory and the pecking order theory. The trade-off theory, introduced by Modigliani and Miller (1963) and refined by Kraus and Litzenberger (1973), posits that firms seek to balance the tax benefits of debt against the potential costs of financial distress. Firms adjust their leverage when deviations from the optimum outweigh the rebalancing costs. In contrast, the pecking order theory, formulated by Myers and Majluf (1984), states that firms prioritise financing sources according to an adverse selection cost hierarchy. They favour internal funds first, followed by debt, and issue equity only as a last resort.

This theory embodies asymmetric information, which means that one group of people, such as managers, has more information about their companies than another group, such as investors.

¹³ F. Modigliani and M. Miller (1963), 'Corporate Income Taxes and the Cost of Capital: A Correction', *American Economic Review*, 1963, pp. 433-43.

Hence, the pecking order theory implies the following:

Firms prefer internal to external financing. Firms adapt their target dividend payout ratios to their investment opportunities. Sticky dividend policies mean that internally generated funds may be more or less than the investment outlays. If these are less, the firm first draws down its cash balances. If external finance is required for capital investment, firms typically issue the safest security first, namely debt. Then, consider hybrid securities such as convertible bonds, and perhaps equity as a last resort.

In the pecking order theory, there is no well-defined target debt-equity mix because there are two types of equity: one at the top of the pecking order and the other at the bottom. Each firm's observed debt ratio explains its cumulative debt requirements and cumulative requirements for external finance.

There is an extensive empirical literature examining the factors that influence firms' choices between debt and equity. See, e.g., the reviews by Frank and Goyal (2009) and Graham and Leary (2011). Although the verdict is still pending, mounting evidence suggests that the trade-off theory is insufficient to explain firms' issuance decisions. Firms' leverage adjustments cannot be explained by rebalancing costs (Eckbo & Kisser, 2021b). Moreover, the consistently negative relationship between profitability and debt issuance, even for firms that already incur rebalancing costs, aligns with the pecking order theory (Eckbo & Kisser, 2021a).

The analysis in Myers and Majluf (1984) does not formally include debt. Instead, they and Myers (1984) formulate an intuitive pecking-order hypothesis, arguing that the informational cost of a debt offering is lower than that of an equity offering. They suggest that because debt is less risky than common stock, the risk of market mispricing is lower, making debt the preferred security.¹⁴

Their contribution lies in explicitly incorporating debt issuance into a theoretical model closely related to the Myers and Majluf (1984) framework. They are the first to formally predict an adverse market reaction to debt issuance, conditional on the

¹⁴ Myers and Majluf (1984) conjecture that “the riskier the instrument issued, the greater the issue's impact on the market value of the firm” but are not “able to prove that this positive relationship is always monotonic” (p. 215).

equilibrium choice of security. Our theoretical analysis demonstrates that firms choose the security that minimises adverse market effects, which can result in optimal issuance of either debt or equity, depending on the circumstances. Thorburn et. al. (2024)¹⁵ test this model using a large sample of debt and equity issues and find robust empirical support for our predictions.

Hanselaar et al. (2019) examine the choice between debt and equity under asymmetric information about the assets in place, using a model similar to that of Myers and Majluf (1984). In a separating equilibrium, high-value firms avoid issuing undervalued securities because the adverse selection cost exceeds the investment opportunity's NPV. Consequently, only firms with values below an endogenously determined threshold issue, leading to an adverse share price reaction upon the announcement of a security offering, whether equity or debt. Issuing firms pool and select the security with the least adverse announcement effect, which may be debt or equity, depending on the project's risk. A second implication of the model is that debt offerings will be associated with less negative announcement returns than equity offerings. This follows from the value of the marginal debt issuer being higher than that of the marginal equity issuer. Hence, the informational cost is lower for debt offerings, conditional on firms' equilibrium debt-equity choice. This implication refines the pecking order hypothesis, which suggests that informational costs are unconditionally lower for debt offerings than for equity offerings.

In the model, when project risk is low, debt investors are likely to be paid from the project, while new equity investors will have a claim on both the project and the existing assets. Conversely, when project risk is high, debt investors are likely to be paid from the assets in place. Therefore, firms prefer to issue equity when project risk is high and debt when project risk is low, which explains the prevalent preference for equity financing among small, research-intensive, high-growth firms.

Thorburn et al. (2024) test the model using a large sample of public debt and equity offerings from 1994 to 2022, comprising listed US firms. They find significantly

¹⁵ Thorburn, et. al. (2024), Conditional Pecking Order, (the Nordic Initiative for Corporate Economics 2024 conference).

negative announcement returns for both debt and equity, with more pronounced adverse market reactions for equity issues, consistent with their model's predictions.

Additionally, Thorburn et al. (2024) employ an endogenous switching regression model to account for endogeneity in the security choice and predict the expected market reaction to the non-issued security. They demonstrate that the announcement return is less harmful for the issued security than for its alternative. The average three-day cumulative abnormal return (CAR) for an equity issue is -2.2%, which would have been -3.0% had the firm issued debt. Similarly, the actual average CAR for a debt issue is -0.8% but would have been -1.9% had the firm instead issued equity. This confirms their model's prediction that issuers select the security with the least adverse market reaction.

Empirical results from Thorburn et al. (2024) are robust in estimating the cumulative asset return (CAR) over longer windows, using three-factor and four-factor models, and across different subsamples. In summary, their conditional pecking order model explains many of the puzzles left by the original pecking order theory, providing a more nuanced understanding of firms' financing preferences.

In essence, Myers and Majluf (1984) suggest that because debt is less risky than common stock, the risk of market mispricing is lower, making debt the preferred security. The contribution of Thorburn et al. (2024) lies in explicitly incorporating debt issuance into a theoretical model closely related to the Myers and Majluf (1984) framework. They are the first to formally predict *an adverse market reaction to debt issuance*, conditional on the equilibrium choice of security. Our theoretical analysis demonstrates that firms choose the security that minimises adverse market effects, which can result in optimal issuance of either debt or equity, depending on the circumstances. Unlike Myers and Majluf (1984), who argue that debt is always the preferred security, Thorburn et al. (2024) model shows that *the preferred security can sometimes be debt and other times equity*.

The model by Thorburn et al. (2024) shows that the asset value of the marginal debt issuer is higher than that of the marginal equity issuer. They test the model's predictions in a sample of 6,100 public debt and equity offerings by US corporations

between 1994 and 2022, sourced from the Securities Data Company (SDC) Platinum's Global New Issues Database. They estimate the three-day cumulative abnormal return (CAR) for each security around the announcement of the issue using a market model. As predicted, the average announcement CAR for debt issues is -0.8%, which is significantly different from zero. Moreover, the average CAR for equity issues is significantly lower at -2.2%, a finding confirmed in cross-sectional regressions. The more negative reaction to equity issues is consistent with their model and existing evidence.

Thorburn et. al. (2024) also examine another implication of our theory: that the market reaction to debt issues, but not to equity issues, becomes more negative as project risk increases. Using the firm's stock-price volatility as a proxy for the riskiness of its investment opportunities, they find a significantly more negative market reaction to debt issues by issuers in the top quartile of stock volatility compared to those in the bottom quartile. However, the average market reaction to equity issue announcements is not significantly different between issuers with high vs. low stock-price volatility. These results, which are confirmed in cross-sectional regressions, align with their model.

Using data on initial public offerings (IPOs) and seasoned equity offerings (SEOs) for 37 countries from 1995 to 2014, Hanselaar et. al. (2019)¹⁶ show that changes in equity issuance are positively related to lagged changes in aggregate local stock market liquidity. This relation is as economically significant as the well-known relation between equity issuance and lagged stock returns. It survives the inclusion of proxies for market timing, capital market conditions, growth prospects, asymmetric information, and investor sentiment. As they posit, changes in liquidity are less relevant for firms under greater financial pressure and for firms in less financially developed countries.

A substantial body of literature demonstrates that aggregate stock market liquidity fluctuates over time within countries (Chordia et al., 2005; Lesmond, 2005). Greater stock market liquidity means that it is easier to sell shares without affecting their price.

¹⁶ Hanselaar, Rogier M., René M. Stulz, and Mathijs A. Van Dijk. "Do firms issue more equity when markets become more liquid?" *Journal of Financial Economics* 133.1 (2019): 64-82.

Authors, therefore, expected that increases in stock market liquidity should be associated with an increase in equity issuance. In the paper, they investigated this hypothesis using a sample of 37 countries from 1995 to 2014. They found strong support for the hypothesis that equity issuance increases following improvements in stock market liquidity. As a firm's shares trade in a less liquid market, investors must be given a greater discount to absorb these shares. Equity issuance is more costly for existing shareholders when a firm's stock is less liquid because an increase in the supply of shares has a greater price impact. As issuance becomes more costly, firms are expected to issue less equity, everything else equal. The liquidity of a firm's common stock can deteriorate due to aggregate liquidity issues or idiosyncratic shocks.

Idiosyncratic liquidity shocks can be caused by shocks to observed or unobserved firm attributes, making it difficult to isolate the impact of liquidity from the impact of shocks to factors that affect liquidity, as well as other firm characteristics. For instance, adverse information about a firm could increase information asymmetry, thereby lowering liquidity. Since one would expect an increase in information asymmetry to make it more expensive for a firm to issue equity, identifying the liquidity effect on equity issuance when liquidity changes due to information asymmetry would be challenging.

An additional complicating factor is that most individual firms issue equity rarely, so tests at the firm level are unlikely to have much power. Aggregate liquidity could proxy for the market's general capacity to absorb new shares. Reverse causation is far less of a concern, since new issues tend to represent a small fraction of the overall market. Earlier papers have investigated equity issuance globally, including Henderson et al. (2006)¹⁷ And Kim and Weisbach (2008).

Hanselaar et. al. (2019) show that issuance is more strongly related to liquidity in more financially developed markets, consistent with the view that firms can issue equity more rapidly in those countries. In contrast, the relationship between issuance and liquidity is weaker for firms incurring losses, which suggests that in circumstances where issuing equity is a matter of greater urgency, liquidity considerations play a lesser role. Furthermore, they show that accounting for variation

¹⁷ Haung, R., Ritter, J.R., 2016, Corporate cash shortfalls and financing decisions. Unpublished working paper, University of Florida.

in liquidity not only improves explanatory power for issuance variation in the sample but also enhances out-of-sample predictive power.

During the 1990-2001 period, firms raised about \$25.3 trillion of new capital, including \$4.9 trillion from abroad. International debt issuances are substantially more common than equity issuances, with debt (equity) issues accounting for 87% (9%) of all securities issued internationally, and about 20% (12%) of all public debt issuances. Market timing considerations appear to be important in security issuance decisions in most countries.¹⁸

2.3.5. Trade-off theory (TOT)

The trade-off theory proposed by Kraus & Litzenberger (1973) is an extension of the MM hypothesis (MM, 1963) with corporate taxes, which balance the costs of debt in the form of financial distress costs (FDC) against the benefits of debt in the form of the present value of interest tax shield (PVITS). The direct costs of financial distress include expenses associated with the firm's insolvency. Due to the conflicting interests of debt holders and shareholders, firms may experience a delay in the liquidation process, which can negatively impact the realisable value of assets.¹⁹ The firm may also be required to sell its assets at distressed prices, which may be significantly lower than their current values. Insolvency proceedings also entail high legal and administrative costs. Even without insolvency proceedings, numerous indirect costs are associated with the actions of employees, customers and suppliers resulting from financial distress caused by higher financial leverage ratios.

With rising costs of financial distress, debt holders may expect higher returns, which may negatively affect the market value of equity shares, thereby adversely affecting the value of the levered firm. The firms try to balance the PVITS and FDC by maximising the positive difference between these two.

$$V_l = V_u + \text{PVITS} - \text{FDC}$$

¹⁸ Henderson, (2006), World Markets for Raising New Capital, Journal of Financial Economics, Vol. 82, No. 1, 2006, pp.63-101.

¹⁹ Developments in the liquidation and bankruptcy process of Jet Airways are a case in point to be noted.

The trade-off theory explains the observed variations in the leverage of firms, with the size of the firm, profitability, tangibility of assets, and operating cash flows positively affecting the leverage ratio of the firm, while growth, product uniqueness, non-debt tax shield, bankruptcy and business risk negatively affect the leverage ratio of the firm.

The trade-off theory suggests that a firm will borrow up to the point where the marginal value of tax shields on additional debt is offset by the increase in the present value of potential costs associated with the company's financial distress. Financial distress refers to the costs associated with bankruptcy or reorganisation.

A firm's optimal debt ratio is usually viewed as determined by a trade-off of the costs and benefits of borrowing, *ceteris paribus* (or holding the firm's assets and investment plans constant).

Value of firm = *Value if all-equity finance* + PV (Costs of Financial Distress) + PV (tax shield)

The tax shield's present value (PV) initially increases as the firm borrows more. At moderate debt levels, the probability of financial distress is negligible, so the present value (PV) of the costs of financial distress is small, and tax advantages prevail. However, beyond a certain point of accumulating more debt, the probability of financial distress increases, and the costs of distress begin to reduce the firm's value. Furthermore, if the firm cannot be certain of profiting from the tax benefit, the tax advantage of additional debt is likely to diminish and eventually disappear.

However, the trade-off theory cannot explain why profitable firms within the industry have lower debt ratios or no debt at all. One plausible explanation for such cases is the consistently high profitability, which may not be viewed as a lucrative option for borrowing. This indicates the entrepreneurial characteristics of these captains of industry and their constant faith in the high profitability of the ventures they undertake. The pecking order theory provides some explanations.

The static trade-off model of capital structure suggests that firms choose their capital structures to balance the benefits of debt financing (such as corporate tax savings and the reduction of agency conflicts) with the direct and indirect costs of financial distress. Several papers have tested this view, including Bradley et al. (1984), Titman

and Wessels (1988), Rajan and Zingales (1995), Fama and French (2002) and the review of Frank and Goyal (2003). Can the trade-off theory explain how companies behave in their ‘corporate finance decisions’? Yes and no. Yes, because the theory does explain many industry differences in capital structure. As mentioned in many earlier studies, high-tech firms, whose assets are risky and intangible, typically use relatively little debt. In contrast, heavy fixed-asset companies, such as airlines, borrow heavily because their assets are tangible and secure. And no, because the theory does not explain why some of the most successful companies thrive with little debt. A prime example of such firms is the highly profitable pharmaceutical company Merck, which is almost entirely equity financed. Under the trade-off theory, high profits should mean a greater debt-servicing capacity and more taxable income to shield, and thus should result in a higher target debt ratio. As we have noticed in censored and truncated variables, there is a limited range of values where the causality may be validated.

Generally, empirical evidence suggests that business risk hurts corporate borrowing decisions. However, there are conflicting conclusions on the impact of other firm-specific variables. While Kim and Sorensen (1986) provided evidence on the negative relationship between non-debt tax shields and leverage, Bradley et al. (1984), Titman and Wessels (1988), and Homaifar et al. (1994) failed to provide such support. There are also varied conclusions on the relationship between size and leverage. Kim and Sorensen (1986) and Chung (1993) show no systematic association between firm size and capital structure. In contrast, Homaifar et al. (1994) and Titman and Wessels (1988) reported results which are consistent with the notion that larger firms have higher debt ratios. Finally, empirical evidence exists on the antagonistic relationship between profitability and debt ratios (the pecking order model of capital structure). For example, the findings of Kester (1986), Titman and Wessels (1988), and Rajan and Zingales (1995) lend support to this relationship. However, Long and Malitz (1985) did not find such a relation between leverage and profitability.

Myers and Majluf’s (1984) and Myer’s (1984) pecking order theory, although not too distinct from the trade-off theory, differs only in its views on which market frictions are most relevant. This theory suggests that the adverse selection costs of issuing equity are large enough to render other costs and benefits of debt and equity less important. While Myers and Majluf demonstrated that the theory is most likely

relevant for firms where the value of growth opportunities is low relative to existing assets, Leary and Roberts (2010) found that it struggles to accurately predict issuance decisions. Although the pecking order view may be a functional, conditional theory (Myers, 2001), as with the trade-off view, it does not explain many financing decisions.

Frank and Goyal (2003) tested the pecking order theory on a broad cross-section of publicly traded US firms from 1971 to 98. The authors find that the internal financing was insufficient to cover average investment spending, contrary to what is often suggested. External financing is heavily utilised, and debt financing does not dominate equity financing in terms of size. Net equity issues track the financing deficit quite closely, while net debt does not do so. These findings are surprising from the perspective of the pecking order theory. Nonetheless, the theory competes with the conventional leverage regressions.

Bradley et al. (1983) examined the modern balancing theory of optimal capital structure, which incorporates positive personal taxes on equity and bond income, expected costs of financial distress (bankruptcy costs and agency costs) and positive non-debt tax shields. They find that optimal firm leverage is inversely related to the expected costs of financial distress and the amount of non-debt tax shields. A simulation shows that if the costs of financial distress are significant, optimal firm leverage is related inversely to the variability of firm earnings. The authors investigate the cross-sectional behaviour of 20-year average firm leverage ratio of 851 US firms and find that: first, there exists a strong industry influence across these firm leverage ratios; second, the cross-sectional regressions on industry dummy variables explained 54 % of variation in firm leverage ratios; and third, the volatility of earnings of firms is an important, inverse determinant of firm leverage and helped explain both inter- and intra-industry variations in firm leverage ratios.

Graham and Leary (2011, p. 310) identify several shortcomings of traditional models in their review of empirical research on capital structure. In general, the explanations of these limitations differ in their assumptions and implications about the nature of the traditional models' problems. For example, the problem lies not in the models themselves but in the empirical measures of leverage and proxies for firm characteristics, or biased estimates of model parameters. Also, although the general

framework of a given model is appropriate, the list of relevant market frictions is typically incomplete. Other research suggests that the correct frictions have been identified, but the implications of these frictions for financial policies are incomplete without additional insights from dynamic considerations. Finally, another possible explanation is that the impact of model leverage changes on firm value may be small, resulting in neutral mutations in the observed data.

Parsons and Titman (2008) offer a synthesis of three themes in their review of the empirical capital structure literature. For the first theme, they examine the evidence on the cross-sectional determinants of capital structure. This literature identifies and discusses the characteristics of firms that tend to be associated with different debt ratios. For the second theme, they review the literature that examines changes in capital structure. The papers examine factors that deviate from their target capital structure. The papers examine factors that cause firms to deviate from their target capital structures, as well as the extent to which future financing choices bring them back toward their targets. Finally, they discuss a set of studies that explore the consequences of leverage, rather than its determinants. These studies focused on the feedback from financing to real decisions; for example, how a firm's financing choices influence its incentives to invest in its workers, price its products, form relationships with suppliers, or compete aggressively with competitors.

2.3.6. Trade-off versus pecking order

Since the pecking order theory delves into different contexts, deciding what comes first – internal, debt or equity - keeping in mind the asymmetry of information and the objective of maximising the gains of existing shareholders rather than the potential/future shareholders, its considerations are not precisely comparable with those of trade-off theory. In trade-off theory, the concern is finding the balance point of scale, which involves weighing not only the minds of borrowers but also those of lenders. This balance point is more difficult to determine if the borrowed money is mobilised from capital markets rather than from a select group of lenders. In the case of a select group of lenders, not only is the amount and the cost of capital negotiable, but there is also scope for collusion among the agents of lending and borrowing entities, depending on the laxity or strictness of the regulatory system. So, the main

theoretical challenge is how to juxtapose the two theoretical arguments together and whether this kind of juxtaposition is possible in all situations, or in some cases, one argument would be more appropriate, and in some other cases, the other argument, and in still other cases, neither of the arguments is doing the job. Now, if we reach that kind of stage, we are virtually challenging the predictive power of arguments – trade-off and pecking order both or maybe one of these, not sure which one – and that is a tall order to achieve²⁰.

Dimensions over and above individual firms

There are aspects of business that operate at an aggregate level rather than at the level of individual firms. Policy, technology, or labour conditions, such as the availability of a trained workforce, may be dimensions that influence an industry as a whole. Likewise, financial market conditions, such as currency exchange rates, interest rates, and growth opportunities, may influence the entire economy.

2.3.7. The free cash flow theory

Agency conflicts: between shareholders and managers, and between shareholders and debtholders.

The former conflict arises because managers will act in their own interest and seek perquisites or additional benefits in the form of job security and higher salaries, as well as attempt to capture cash flows by retaining them in the firm instead of disbursing them, even when there are no growth opportunities.

“The problem is how to motivate managers to disgorge the cash rather than investing it at a rate of return below the cost of capital or wasting it in organisational inefficiencies”²¹. Hence, the greater the discretionary amount available to a manager, the greater the likelihood is that he will use it for perquisites for himself (this is also referred to as *the overinvestment problem*). The solution can be debt, which forces the firm to pay out cash. As a result, a manager’s ability to promote self-interest is

²⁰ Modigliani and Miller were defending their irrelevance theory while replying on aspects relating to dividend as well as tax shields, because these, in their arguments, reflect current frictions in financial/capital markets—MM (1961) on dividends and MM (1963) on corporate income tax.

²¹ Jensen (1986), p. 323.

constrained by the availability of free cash flows and can be further tightened by higher debt financing. Although a high debt ratio can be risky, it can also add value by forcing the firm to walk a tightrope. Consequently, agency problems might be optimally solved through a capital structure decision, such as increasing debt leverage.

Poitevin (1989) proposes a model that uses debt as a signal of efficiency. In a model where an incumbent firm and an entrant compete, a low-cost entrant would prefer to issue debt, while the incumbent and a high-cost entrant would prefer to issue equity. The benefit of debt is that the financial market places a higher value on debt-financed firms since there is an established belief that debt-financed firms are low-cost firms. High-cost entrant firms will not issue debt because the resulting higher probability of bankruptcy makes the cost of misleading the capital market too high.

2.3.8. Signalling theory

Akerlof (1970) first introduced the concept of signalling theory in the context of job and product markets. Spence (1973) developed it into an equilibrium theory. Opposed to the MM hypothesis (MM, 1958), which assumes no information asymmetry, signalling theory is based on recognising information asymmetry. It is fundamentally aimed at reducing the information asymmetry between the two parties. (Spence, 2002). The firm's decision regarding its leverage policy (Ross, 1977) is not only a decision about leverage but also conveys signals about the firm's quality, thereby influencing the quality of the decision regarding capital structure. Ravid and Sarig (1991) demonstrate that firms use leverage decisions to signal the firm's quality. With less cash and internal funds for potential investment opportunities, companies may prefer external equity funding through rights or fresh equity. They also predict that better-quality firms maintain high leverage ratios and pay higher dividends than lower-quality firms. Aharony and Swary (1980) and Masulis (1980) also support the signalling effects of dividend decisions and leverage decisions, documenting positive abnormal returns in the market following firms' increases in dividend payouts or leverage ratios.

2.3.9. Agency cost theory

An agency relationship involves the delegation of decision-making power by the principal to the agent to utilise the resources handed over by the principal in the principal's best interest. A conflict, termed agency conflict, arises because both the principal and the agent are value maximisers. To maximise its value, the principal incurs certain costs – both pecuniary and non-pecuniary – known as agency costs.

$$\text{Agency Costs} = \text{Monitoring costs} + \text{Bonding Costs} + \text{Residual Loss}$$

Jensen and Meckling (1976) identify two types of agency conflicts: one between equity shareholders and management, and another between debt holders and shareholders. The conflict between equity shareholders and management arises because management increases the portion allocated to their compensation or perquisites, thereby reducing the gain that accrues to equity shareholders. (Harris & Raviv, 1991)

Such conflict resolution may involve two measures: Increasing the managerial ownership of the firm (Jensen & Meckling, 1976) and raising the firm's leverage levels, thereby attracting higher monitoring from debt holders (Jensen, 1986). However, the second measure results in a conflict between shareholders and debt holders; a successful project using debt financing benefits equity shareholders more than it benefits debt holders. Furthermore, the cost of a project failure is borne more by debt holders than by equity shareholders. So, leverage is positively related with default probabilities (Harris & Raviv, 1990a) and free cash flow (Jensen, 1986; Stulz, 1990) and negatively related with growth opportunities (Jensen & Meckling, 1976; Stulz, 1990), interest coverage (Harris & Raviv, 1990a) and the probabilities of reorganisation following default (Harris & Raviv, 1990a)

Harris and Raviv (1990) and Stulz (1990) proposed methods to mitigate agency costs associated with manager-shareholder conflict. Specifically, in Harris and Raviv's model, managers are assumed always to want to invest all available funds, even if investors prefer to liquidate the firm. Paying out mitigates the problem in the Harris and Raviv model by giving debtholders options in slow-growth, cash-rich industries. The optimal capital structure, as stated by Harris and Raviv, is outlined through the

following propositions: PROPOSITION 1 (Probability of Reorganisation): If the firm defaults in period one, then the probability that it will reorganise is independent of default costs and debt level, and decreases with increases in the liquidation value function. If the liquidation value does not increase with the current income, then the reorganisation probability increases. Also, under the constant returns to scale assumption, the probability of reorganisation depends on current income and firm scale only through the ratio. PROPOSITION 2 (*Debt Level and Coverage Ratio*): The optimal debt level decreases with increases in default costs and increases with increases in liquidation value. The expected debt coverage ratio moves in the opposite direction of the debt level in response to changes in default costs and liquidation value. Moreover, under the constant returns to scale assumption, the debt level is proportional to the firm's scale, and the expected debt coverage ratio is independent of the firm's scale. PROPOSITION 3 (*Debt Level and Default Probability*): Debt level and default probability move in the same direction with shifts in default costs and liquidation value. Under the constant returns to scale assumption, the default probability is independent of firm scale. PROPOSITION 4 (*Market Value of the Firm*): Market value of the firm increases with increases in liquidation value and decreases with increases in default costs. Under the constant returns to scale assumption, firm value is proportional to firm scale. PROPOSITION 5: (*Market Value of debt is proportional to firm Scale*): Under the constant returns to scale assumption, the probability that debt holders receive full value, promised yield, and debt-to-value ratio are independent of firm scale. The market value of debt is proportional

2.3.10. Corporate life stages (CLS)

Entities and their management have functioning logic that operates at the firm level. The primary motives for equity holders may be survival, growth, and profitability.

Although it may not be easy to trace its origin, financial leverage or capital structure has been a persistent phenomenon for many decades. Not that it does not change, but it does, at times, due to fundamental compulsions. A change in life stage is one such example. Therefore, there are overlapping theoretical explanations when we examine the phenomenon of capital structure and its determinants from different perspectives.

So, what happened in the past stays in the future is the scenario regarding capital structure decisions.

Having reviewed the literature and the general thought process in financing, we now turn our attention to the focus of this study: financing over corporate life stages, as presented in the chapters that follow. This includes not only how we define financing but also how we understand the concept of corporate life stages. In the chapters that follow, we not only review the existing literature but also identify research gaps and attempt to further our understanding. The next chapter is focused explicitly on reviewing the literature in the area, with a particular emphasis on the determinants of capital structure across corporate life stages. There have been two significant contributions in this: Anthony & Ramesh (1992) and Dickinson (2011).

Anthony and Ramesh (1992)

Anthony and Ramesh (1992) identified the life stages of firms using variables such as the year of incorporation, rate of profit, and other relevant factors.

This paper posits that the stock market response to two accounting performance measures—sales growth and capital investment—is a function of the firm's stage in its life cycle. Firms are grouped into various life cycle portfolios using dividend payout, sales growth, and age. As predicted, the empirical results indicate a monotonic decline in the response coefficients of unexpected sales growth and unexpected capital investment from the growth to the stagnant stages. Additional analysis suggests that this relation is not driven by a firm size effect, risk differences, or measurement error in the proxies for performance measures.

Dickinson (2011)

Dickinson (2011) identified ‘corporate life stage’, based on the pattern of cash flows, comprising cash flows from operations, financing and investment, as a control variable.

This study develops a firm life cycle proxy using cash flow patterns. The patterns provide a parsimonious indicator of life cycle stage that is free from distributional assumptions (i.e., uniformity). The proxy identifies differential behaviour in the persistence and convergence patterns of profitability. For example, return on net

operating assets (RNOA) does not revert to the mean (a spread of 7 per cent after five years between mature and declining firms) when examined by life cycle stage, which has implications for growth rates and forecast horizons. Furthermore, determinants of future profitability, such as asset turnover and profit margin, are differentially successful in generating increases in profitability, depending on the stage of the life cycle. Finally, investors do not fully incorporate the information contained in cash flow patterns, and as a result, they undervalue mature firms. The cash flow proxy is a robust tool that has applications in analysis, forecasting, valuation, and as a control variable for future research.

In this article, Ichak Adizes (1979) presents a model for identifying organisational styles. Such a typology is a recurring pattern in a significant number of organisations, and hundreds of executives have verified that it reflects their experience.

The model enables an organisation to foresee the problems it will face as it grows over time. Furthermore, it provides tools for prescribing effective organisational treatments, including organisational therapy and surgery. It is a contingency model in that it presents a framework for prescribing the treatments most likely to be effective depending on the lifecycle stage of the organisation.

While reviewing the relevant literature, the focus is on ‘corporate life stage (s)’, which will be discussed in detail in the next chapter, and on developing a research hypothesis.

2.4. Corporate taxes and capital structure

In this section below, we will discuss various aspects of taxation as they relate to capital structure and debt.

2.4.1. Debt and taxes

In 1963, in continuation of their previous work (MM, 1958), Modigliani and Miller posited that not only is the firm's value independent of its capital structure decision,

but corporate taxes also create an advantage to using debt due to the interest tax shield.

$$\text{Interest Tax Shield} = \text{Interest Paid} * \text{Corporate Tax Rate}$$

Assuming perpetual debt, the present value of the interest tax shield may be calculated as

$$\text{PV of Interest Tax Shield} = \text{Debt} * \text{Corporate Tax Rate}$$

Due to this tax advantage, the firms employing higher debt levels in their capital structure will have greater tax savings, resulting in a higher value of the firms. Therefore, the value of a levered firm will be higher than the value of an unlevered firm.

$$V_l = (\text{NOI}(1-T_c) / K_o) + (\text{Debt} * T_c)$$

where

NOI = Net Operating Income,

T_c = Corporate Tax Rate

However, MM (1963) fails to consider the direct and indirect financial distress costs arising from rising debt levels in the capital structure. It may not be out of place to mention that MM (1963) emphasise that a tax advantage for debt financing does not necessarily mean that corporations should always seek to use the maximum possible amount of debt in their capital structures. For one thing, other forms of financing, notably retained earnings, may still be cheaper in some circumstances when the tax status of investors is taken into account under the personal income tax. More importantly, there are, as we pointed out, limitations imposed by lenders as well as many other dimensions (and kinds of costs) in real-world problems of financial strategy which are not fully comprehended within the framework of static equilibrium models, either our own or those of the traditional variety. These additional considerations, typically grouped under the rubric of "the need for preserving flexibility," will usually imply that corporates maintain *a substantial reserve of untapped borrowing power*. The tax advantage of debt may well tend to lower the

optimal size of that reserve. However, it is hard to believe that the advantages of the size contemplated under our model could justify any substantial reduction, let alone their complete elimination.²²

When firms pay corporate taxes, the income available to investors decreases. Subject to the government's policy, the investors are required to pay personal income taxes on the income they receive from corporates. Miller (1977) examines the impact of corporate and personal income taxes on a firm's value.

$$V_l = V_u + Debt * (1 - (((1-T_c)*(1-T_e))/T_d))$$

where

V_l = Value of the levered firm

V_u = Value of the unlevered firm

T_d = tax rate on personal income received in the form of interest

T_e = tax rate on personal income received in the form of dividends

T_c = Corporate tax rate

In the absence of both corporate and personal taxes, the above equation gets transformed into the original MM hypothesis (MM, 1958) without taxes.

Tax on income and leverage

Interest on debt is fully tax-deductible in India. Additionally, although not directly applicable, the Finance Act of 2020 introduced significant changes to the taxation of dividends in India, effective from Assessment Year 2021-22. The Finance Act of 2013 abolished the dividend distribution tax (DDT), shifting tax liability from companies to shareholders. It has implications for retained earnings, becoming a more important source of financing.

The Companies Act of 2013 includes provisions that relate to financial statements, loans, and investments, which are related to financial leverage:

²² MM (1963), p. 442.

2.4.2. Financial statements

Section 129 of the Companies Act, 2013, requires companies to classify short-term borrowings, including loans, deposits, and advances, from related parties as such. It also requires companies to sub-classify borrowings as secured or unsecured, and to disclose the nature of the security.

2.4.3. Loans and investments

Section 186 of the Companies Act, 2013, states that companies can make investments through more than two layers of investment companies. However, companies cannot directly or indirectly give loans to any person or body corporate; Provide a guarantee or security in connection with a loan to any other person or body corporate; Acquire securities of any other body corporate that exceed 60% of its paid-up share capital, free reserves, and securities premium account.

Financial leverage is the use of borrowed capital or debt to finance a project or investment. It is a concept that both investors and companies use. Investors use leverage to increase the returns on their investments, while companies use it to finance their assets.

The Companies Act of 1956 also provided the legal framework for corporate entities in India, including provisions to protect the interests of the public, shareholders, creditors, and employees. Here are some aspects of the Companies Act of 1956 that relate to financial leverage and the protection of investors: Inspection of books of accounts: The Central Government can inspect a company's books of accounts to ensure that the company is operating following the Act; Special audits: The Central Government can direct a special audit of a company's affairs; **Prosecution:** The Central Government can prosecute a company for violating the Act; **Companies (Acceptance of Deposits) Rules:** These rules require companies to publish their financial accounts when inviting deposits; **Repayment of deposits:** The Company Law Board can direct a company to repay deposits if it fails to do so under the terms and conditions.

Debt-to-equity ratio: A debt-to-equity ratio of not more than 3:1 may be one of the conditions for obtaining financial assistance for a leveraged buyout (LBO). Empirical evidence on the leverage ratio has been contradictory among different authors. Miller (1977) believes that capital structures have shown so little change over time.²³ On the other hand, Adair Turner (2023)²⁴ shows that corporate debt and household debt levels were counterbalanced after the 2008 global financial crisis (GFC).

Economies worldwide implement similar rules and regulations. Since the study focuses on Indian firms, provisions related to the Indian legal system have been specifically highlighted.

2.5. Methodologies used in research on determinants of capital structure

Several methodologies may be discussed in detail:

2.5.1. Limited dependent variable linear, multiple discriminant analysis

Categorical dependent variable: categorical variable models – the section focuses on the dependent variable only as a categorical variable, emphasising that R-square or adjusted R-square has no meaning, instead goodness of fit measures like success rate (percentage) or pseudo-R-square, also known as McFadden's R-square²⁵ are the right indicators.

²³ Miller, Merton H (1977), Debt and Taxes, *Journal of Finance*, Vol 32 No 2, May 1977, pp. 261-275. "The debt/asset ratio of the typical non-financial corporation in the 1950's was little different from that of the 1920s even though tax rates had quintupled-from 10 and 11 percent in the 1920's to 52 percent in the 1950's The remarkable stability of corporate debt ratios in the face of huge increases in tax rates was noted by many other writers in this period. See, e.g., Arnold W. Sametz. "Trends in the Volume and Composition of Equity Finance," *Journal of Finance*, Vol. 19, No. 3 (September 1964), 450-469.

²⁴ Turner, Adair (2023), *Between Debt and the Devil: Money, Credit, and Fixing Global Finance*. (Princeton University Press, 2023).

²⁵ Pseudo R^2 is defined as equal to $(1 - (LLF/LLF_0))$ where LLF is the maximum value of the log-likelihood function for the logit and probit model and LLF_0 is the value of the log-likelihood function for a restricted model where all the slope parameters are set to zero (that is, the model has only an intercept). Pseudo- R^2 will have a value of zero for the restricted model. For the limited-dependent variables or categorical variables, in this case, we can use (multinomial) probability models, such as logit and probit. Estimation of such models is done via the maximum likelihood approach. For the logit and probit models, this means that they took only two values, 0 or 1. For multinomial models, the dependent variable takes various values (arbitrarily set) to reflect each possible characteristic the variable can take.

Censored and truncated dependent variables

Panel analysis with discussion on pooled, seemingly unrelated regression, and fixed-effects and random-effects models.

Panel data, also known as longitudinal analysis, combines time-series and cross-sectional data. A panel (of data) uses the same entities (cross-section, i) and computes some quantity about them over time (the time series, T). A panel can be described as

$$Y_{it} = \textit{Alpha} + \textit{Beta} * X_{it} + U_{it}$$

where Y_{it} is the dependent variable, *Alpha* is the intercept term, *Beta* is a $k \times 1$ vector of parameters to be estimated on the explanatory variables, and X_{it} is a $1 \times k$ vector of observations on the explanatory variables, $t = 1, \dots, T$; $i = 1, \dots, N$.

What are some of the estimation methods?

2.5.2. Pooled regression (using OLS)

It involves estimating a single equation for all the data after stacking all entities and explanatory variables on top of each other. A shortcoming is that it implicitly assumes the average values of the variables and their relationship are constant over time and across all cross-sectional units in the sample.

2.5.3. Seemingly Unrelated Regression (SUR) framework

Zellner (1962) proposed that even though the dependent variables seem unrelated across the equations at first sight, a more careful consideration would allow us to conclude that they are related after all. Unfortunately, SUR suffers from two limitations:

First, SUR can be used when the number of time-series observations, T , per cross-section unit, X , is at least as large as the total number of such units. For example, if you are studying 371 private firms, you should have at least a time series for 371 years or periods.

Second, the total number of parameters estimated is enormous, resulting in a large variance-covariance matrix of errors, which can be challenging to estimate accurately.

A third approach to estimating panel data is a setup that corrects both the large data estimation problem and the lack of dynamic changes in the parameters. This approach is widely used and entails two broad classes of panel estimators: Fixed effect and random effect

2.5.4. The fixed-effects model

Fixed effects regression is a statistical method used in panel data analysis to account for unobserved individual-specific characteristics that may be correlated with the independent variables, potentially leading to biased results in standard regression models. By including fixed effects, the model effectively controls for these unobserved, time-invariant factors, allowing for a more accurate estimation of the causal effect of the independent variables on the dependent variable.

2.5.5. The random-effects model

The random-effects model is an alternative to the fixed-effects model. Random effects regression is a statistical method used when analysing clustered or panel data, where observations within groups are not independent. It assumes that there are random, unobserved effects specific to each group, which are incorporated into the model. This allows for the estimation of group-specific intercepts or slopes, acknowledging that observations within the same group may be correlated.

The discussion on capital structure methodology concludes with an examination of econometric issues in these methodologies. This is followed by empirical evidence on capital structure decisions.

Harris and Raviv (1990) proposed a theory of capital structure based on the idea that debt enables investors to discipline and monitor management. In their model, investors use information about the firm's prospects to decide whether to liquidate the firm or continue current operations. The authors assume that managers are reluctant to liquidate the firm under any circumstances and are unwilling to provide detailed

information to investors that could result in such an outcome. Investors gather information from the firm's ability to make payments and from the potential costs of a costly investigation in the event of default. Debtholders use their legal rights to force management to provide information and implement the resulting liquidation decision. The optimal amount of debt is determined by trading off the value of information and opportunities for disciplining management against the probability of incurring investigation costs. Consequently, stockholders will design debt payments (or capital structure over time) to exploit the ability of debt to generate useful information.

Titman and Wessels (1988) extend the empirical work on capital structure theory in three ways, in addition to their criticism of the inappropriate use of attributes in capital structure regressions. First, they extend the range of theoretical determinants of capital structure by examining some recent theories that have not been analysed empirically. Some of these theories have been presented in the previous section. Second, they analyse separate measures of short-term, long-term and convertible debt rather than an aggregate measure of total debt. Moreover, third, they use a new technique which explicitly recognises and mitigates the measurement problems. This technique, which is an extension of the factor-analytic approach to measuring unobserved or latent variables, is known as linear structuring modelling. Following Titman and Wessels (p. 2), the method assumes that, although the relevant attributes are not directly observable, they can observe several indicator variables that are linear functions of one or more attributes and a random error term. This specification directly resembles the Arbitrage Pricing Theory (APT) and is very similar to the procedure used by Roll and Ross (1980) to test the APT.

Other studies in the literature have focused on the determinants of the speed of adjustment to financial targets, providing more direct evidence that firms adjust toward a target debt ratio. Taggart (1977) provides evidence that the speed of adjustment to long-term capital targets is relatively slow and that short-term debt plays a significant role in the adjustment process. Marsh (1982), using a logistic specification, analysed the choice of financing instrument of companies and argues that the choice depends on the difference between the company's current and target debt ratios. His results suggest that companies try to maintain their long-term debt

levels, although they deviate from these targets in the short run in response to capital market conditions.

Jalilvand and Harris (1984) examine the determinants of the speed of adjustment to long-term financial targets, allowing the speed of adjustment to vary across companies and over time. Their results imply that firm size, interest rates, and stock price levels affect the speed of adjustment.

There is also research on the international front, mainly for UK firms. Bennet and Donnelly (1993) examined cross-sectional determinants of leverage decisions among non-financial UK firms. He found that non-debt tax shields, asset structure, size and profitability exerted a substantial impact on the capital structure choice of firms. Lasfer (1995) examined the impact of the corporate tax and agency costs on firms' borrowing choices. He presented evidence that firms with limited growth opportunities tend to have higher levels of debt in their capital structure. Additionally, firms more likely to experience free cash flow issues often have lower debt ratios. Furthermore, corporate tax appears to have a negligible impact on firms' short-term capital structure choices. Walsh and Ryan (1997) find that agency and tax considerations are significant in determining debt and equity decisions of the UK firms. Ozkan (2001) also examines British firms and finds that they target leverage ratios, and they adjust to the target ratio relatively quickly. This suggests that the costs associated with deviating from their target ratios and the costs of adjustment are important. It also finds that the current liquidity and profitability of firms exert a negative impact on their borrowing decisions. However, a positive relationship exists between past profitability and the debt ratio. Finally, firm-specific variables which appeared to influence leverage decisions were non-debt tax shields and growth opportunities of firms. However, there was limited evidence that firm size exerts an impact on capital structure decisions.

Korajczyk and Levy (2003) examined the impact of macroeconomic conditions and financial constraints on the choice of capital structure. Firms facing financial constraints do not choose capital structure in the same manner as unconstrained firms. Similarly, time variation in macroeconomic conditions, such as changes in the relative pricing of asset classes, can lead a given firm to choose a different capital structure at different

points in time, all else being equal. Their methodological approach was similar to that of Hovakimian et al. (2001), which examines the relationship among firm-specific variables, target leverage, and the choice of issue. Korajczyk and Levy (2003, p. 76) split their sample into financially constrained and financially unconstrained firms. They define financially constrained firms as those that lack sufficient cash to undertake investment opportunities and face severe agency costs when accessing financial markets. Moreover, they estimate the relationship between firms' debt ratios and both firm-specific variables and macroeconomic conditions. Using the fitted values of this relationship to estimate firms' target capital structures, they then investigate the relation between security issuances/repurchases, the deviation from target leverage, and both firm-specific and macroeconomic variables. Empirically, the relation between firm-specific variables and target leverage is consistent with some elements of both the pecking order and the trade-off theories of capital structure. However, the relation is also inconsistent with some elements of each theory. For instance, larger firms and those with more tangible assets tend to have higher leverage, and firms with unique assets tend to have lower leverage. Consistent with the trade-off theory, firms with large depreciation tax shields have lower target leverage. However, the antagonistic relation between operating income and leverage seems consistent with a pecking order theory, particularly for unconstrained firms (Korajczyk and Levy, 2003, p. 77).

Margaritis and Psillaki (2010) examined the relationship between capital structure, ownership structure and firm performance using a sample of French manufacturing firms. Applying nonparametric data envelopment analysis methods to measure firm efficiency empirically, they sought to determine whether more efficient firms prefer debt in their capital structure. Further, employing quantile regressions, they tested the effect of efficiency on leverage and thus the empirical validity of the two competing hypotheses across different capital structure choices, namely the efficiency-risk and franchise value hypotheses. They find that the effect of efficiency on leverage was positive in the low to high ranges of the leverage distribution, supporting the efficiency-risk hypotheses. They also found that more concentrated ownership was generally associated with more debt in the capital structure. Finally, they test the direct relationship from leverage to efficiency stipulated by the Jensen and Meckling

(1976) agency cost model and find support for the core prediction of the hypotheses in that higher leverage is associated with improved efficiency over the entire range of observed data.

Leland (1994) examined the optimal capital structure and the pricing of debt with credit risk. His assumption of infinite life debt, although it offered closed-form solutions for debt values and equity values with endogenous bankruptcy, is restrictive. Firms must choose both the maturity and the amount of debt. Leland and Toft (1996) extended Leland's (1994) results to examine the effect of debt maturity on bond prices, credit spreads, and the optimal amount of debt. Leland and Toft (1996) demonstrate that longer-term debt more effectively leverages tax advantages, as bankruptcy typically occurs at lower asset values. However, longer-term debt also creates greater agency costs by providing incentives for equity holders to increase firm risk through asset substitution. Their findings highlight how the twin dimensions of optimal capital structure, amount and maturity represent a trade-off between tax advantages, bankruptcy costs and agency costs.

2.6. The dynamic models

A dynamic regression model incorporates the time-dependent nature of data into the regression process, allowing for the modelling of how changes in predictor variables affect the dependent variable over time. These models go beyond standard regression by considering the time-series dynamics of both the predictor and dependent variables.

2.7. Chapter summary

This chapter reviews the extensive literature in the field of financing to highlight why this study focuses on the life stages of firms when studying capital structure. That is where all the upcoming chapters are organised. To start with, the literature review on corporate life stages and capital structure aims to identify research gaps (Chapter 3), outline the data and methodology (Chapter 4), and present the study's outcomes (Chapters 5 to 8). The summary, recommendations, and further scope of research are also provided (Chapter 9).

Chapter 3
Review of Literature

CHAPTER 3

REVIEW OF LITERATURE

In this chapter, the study outlines the contours of research in the domain of capital structure, a relevant decision in corporate finance, and its determinants. We will first briefly analyse early capital structure theories and then examine more recent work in the domain. However, since we focus on corporate finance primarily in terms of corporate life stages, we will confine our discussion to this objective.

3.1. Introduction

Capital structure is a significant decision of corporate finance, impacting the way money is mobilised for the long-term functioning of the business firm.

Early theories of capital structure

From the time when financial markets operated with a huge asymmetry of information, making it difficult to know how the left arm was behaving, the captains of industry used to rule the roost. Capital in search of returns

There are so many theories that are based on the objective of stockholder-wealth maximisation that deserve consideration:

Classical or traditional, developed systematically by Ezra Solomon, postulates that there is a range of optimal capital structures, that is, a range of debt/equity ratios over which the discount rate (or the average cost of capital) attains its minimum value, resulting in the maximisation of the market value of the firm (over that range). All these examine the question as to whether there is a capital structure that maximises the value of the firm to the existing (current) stock (share) holders.

The theory developed by Modigliani and Miller (MM), under the assumption of perfect capital markets and no corporate (or other) taxes, postulates that the value of the firm, and consequently, the wealth positions of stockholders, is not affected by the capital structure (type of financing).

As a modification, it takes into account corporate taxes. It postulates that a firm should use as much debt as possible to maximise its value and the wealth position of its stockholders.

Managerial theory: How managers in practice decide the proportion of debt and equity in the capital structure of the firm.

Trade-off theory and pecking order theory are two significant theoretical arguments which will be deliberated upon.

Corporate life stages (CLS): Financial life stages of the corporates bring in the dimensions of trade-off, pecking order, managerial aspects,

There are theories, but no law of financing. No more a decision, just a law with maybe some exceptions. Law could be there in the long run. However, we have a long way to go. It is generally not feasible to incorporate all the factors that bear on the decision, in this case, the financing decision, in the study that examines the phenomenon. Therefore, we end up with various theories: MM, TOT, POT, and CLS. Some studies propose theories, and others replicate and extend those theories with new data, countries, industries, and other explanatory variables. At times, these studies yield something that could be a completely new theory, such as a trade-off or pecking order. Interwoven with others, such as information asymmetry, corporate life stages, and trade-off/pecking order.

For a law of financing, it is still relatively early. The phenomenon remains too complex to be stated as a law, or perhaps it is simply not possible, given the dynamic nature of the decision-making context. There are numerous dimensions, and the data is still scarce. There are few studies in the public domain, or in both public and private domains, from which to draw such conclusions.

When a firm borrows (i.e., takes out a long-term loan), it leverages its existing resources. The equity that the firm holds serves as a capital security in the event of a default. The firm's profitability is a guarantee that interest will be paid on time. It is also a source of attraction to gain returns on equity capital, even if the firm does not have additional resources to deploy as equity. However, it is also a risk because, due

to the turn of the tide, profitability may decline, and the risk of default may arise; therefore, one must opt for a loan carefully and within limits. Lenders are equally cautious to watch for such circumstances. Alternatively, the lenders may run the risk of losing their debt capital, not to mention returns on such lent money.

Any serious contemporary literature on corporate financing begins with an exposition of the award-winning theorem propounded by Modigliani and Miller in 1958 regarding the irrelevance of the capital structure to the value of the firm. Under the strict assumptions of perfectly efficient capital markets, which are difficult to operate, the capital structure becomes an instrument that enhances or reduces the value of the firm. Furthermore, capital structure decisions have been of significance in practice and an important area of study in corporate finance.

The literature also examines the calculation of the cost of capital, specifically the Weighted Average Cost of Capital (WACC), with or without considering taxes. Since valuation is derived from market parameters and cost of capital of the firm, or WACC or r_a is based on overall value and equity cost, or r_e is a derived measure, or likewise, debt cost or r_d is also a derived measure, the capital structure decision does not affect the value of the firm. Since valuation is based on discounting future cash flows to present value, under perfectly efficient capital markets, implying no taxes, the value of a firm cannot be affected by capital structure decisions.

According to MM, and practically speaking, debt has a prior claim on a firm's assets and earnings. The following paragraphs discuss several theories of capital structure. These paragraphs present several empirical (econometric) methodologies for understanding how capital structure is implemented and tested, including multivariate discriminant analysis, categorical variable models, panel analysis, and Altman's Z-score methodology. Following these paragraphs on capital structure, a discussion ensues on some notable empirical evidence. Additionally, the discussion of conflicting or overlapping concepts concerning maximisation objectives or the guiding goal is fascinating and important, but is kept separate from the discussion on theories. That level of discussion on maximisation objectives has its intricacies at a philosophical level.

Following Harris and Raviv (1990), capital structure theories can be classified into various categories, including those emanating from agency costs (conflicts) or asymmetric information. Alternatively, these could be those emerging from the interaction of capital structure with competition in the product market and/or the product's inputs, or those that derive from corporate control considerations.

Following Myers (1984), there are two ways of thinking about capital structure. The first is the static trade-off framework, in which the firm is viewed as setting a target D/V ratio and gradually moving toward it. The second is the traditional pecking order framework, in which the firm prefers internal to external financing when issuing securities.

The chapter discusses the following theories of capital structure in detail: the trade-off theory, the pecking order theory, the free cash flow theory, and other related theories.

3.2. Corporate life stages and capital structure

Prior studies have examined the corporate life cycle in various aspects of corporate finance, including dividend payout policy (DeAngelo et al., 2006; Coulton & Ruddock, 2011), seasoned equity offerings (SEOs) (DeAngelo et al., 2010), share repurchases (Liang et al., 2013) and mergers and acquisitions (M&As) (Owen & Yawson, 2010; Arikan & Stulz, 2016). While prior studies use different indicators, such as firm age (Liang et al., 2013) and firm age from IPO (Arikan & Stulz, 2016), in order to classify corporate life cycle stages of firms, this study follows Anthony and Ramesh's (1992) and Dickinson's (2011) procedure to determine a firm's corporate life cycle stages. The advantages of Anthony and Ramesh's (1992) procedure consider not only firm age but also financial variables to classify corporate life cycle stages of firms. As firms within different industries may be at various stages of their corporate life cycle, it would be helpful to consider financial variables in determining the stage of a firm's corporate life cycle. Controlling for financial variables can also enable the current study to additionally consider the size effect in determining corporate life cycle stages of firms.

Anthony and Ramesh (1992) argue that firms in the early stages of the corporate life cycle generally experience higher sales growth. Growth firms have lower dividend

payout ratios because they may have more opportunities to undertake projects with positive net present value. In addition, growth firms are more likely to invest funds in fixed assets, such as plant and equipment. Hence, Anthony and Ramesh (1992) use four variables to identify the corporate life cycle, where life cycle descriptors include annual dividend as a percentage of income (DP), percentage sales growth (SG), capital expenditure as a percentage of total value to the firm (CEV), and firm age (AGE).

Having set the tone of the study, this chapter provides a comprehensive review of the research dimensions in corporate financing across corporate life stages.

3.2.1. Corporate life cycle

In the published literature, various measures are used to characterise the stages of the business life cycle, including firm age, size, growth, strategies, flexibility and controllability (La Rocca *et al.*, 2011). The classification of life cycle stages reflects the variation of business growth and the adaptability of businesses to the competitive environment (Bulan & Yan, 2009). Cash flow patterns have been identified as a valid measure of life cycle stages, reflecting the interaction among resource allocation, operating capacity, and business strategies (Thanatawee, 2011).

Measuring the business life cycle using cash flow patterns enables us to capture non-sequential transitions between stages, which sequential proxies, such as firm age, cannot measure. Dickinson (2011) provides a detailed classification of business life cycle stages using cash flow patterns and demonstrates its reliability in predicting business performance.

There is considerable empirical evidence on the impacts of life cycle stages on various aspects of business behaviour, performance and decision-making, such as foreign direct investment (L demand on financial products (Berger *et.al.*, 1998), market selection (Bellone *et al.*, 2008), cost of finance (Kim *et al.*, 2014), strategy-making (Verreynne & Meyer, 2010), dividend policy (DeAngelo *et al.*, 2006) and accounting practice (Kallunki & Silvola, 2008).

Life cycle effects have also been found in the investigation of the impacts of financial performance on environmental policy (Elsayed & Paton, 2009), the impacts of ownership structure on corporate performance (Liang *et al.*, 2011) and credit rating migration risks (Fei *et al.*, 2012), but few have examined transitions in the business life cycle.

	1 Intro	2 Growth	3 Mature	4 Shake- out	5 Shake- out	6 Shake- out	7 Decline	8 Decline
Predicted Sign								
Cash flows from opera- tions	–	+	+	–	+	+	–	–
Cash flows from in- vesting	–	–	–	–	+	+	+	+
Cash flows from finan- cing	+	+	–	–	+	–	+	–

Source: Dickinson (2011)

3.2.2. Life cycle theory of capital structure

Firm life cycle stages are distinct phases that result from both internal (e.g., strategy choice, financial resources, and managerial ability) and external factors (e.g., competitive environment and macroeconomic factors) (Dickinson, 2011). Dickinson states firms are comprised of multiple products and services, each with its own product life cycle stage. However, those product life cycle stages can be aggregated to produce a life cycle stage mapping at the firm level. Unlike product life cycles, firm life cycles do not need to progress linearly through the introduction, growth, maturity, and decline stages. It is optimal for a firm to recycle between the growth and mature stages, where profits are maximised and risk is presumably lower than it would be in the tails (e.g., introduction and decline).

Dickinson (2011) measures a firm's life cycle using cash flow patterns from the statement of cash flows. Furthermore, she documents that each life cycle stage is present in nearly all industries, thereby alleviating concerns that the firm life cycle is

driven by industry membership. She also demonstrates the superiority of the cash flow pattern life cycle measure over other proxies used in prior literature, such as age, size, sales growth, capital expenditures, research and development, and dividend payments (e.g., Anthony & Ramesh, 1992; Black, 1998).

3.2.3. Literature covering firms from other countries

Castro et. al. (2015) analyse the effect of a firm's life cycle stages on the capital structure in tech versus non-tech firms using a broad sample of public companies from Europe. An innovative approach based on operating, investing, and financing cash flows allows for the analysis of differences in leverage, specifying the differential role of significant drivers of the capital structure across stages in both sectors. The results of the study point to the information asymmetry factor posed by the pecking order as the predominant driver behind the differences in the effect of intangible assets and growth opportunities for tech firms in some stages, mainly maturity. Frank and Goyal's (2003) test of the pecking order theory confirms the lower use of debt by tech firms during all life cycle stages. In addition, they find that the results obtained for tech firms are primarily attributable to the behaviour of high-tech firms with the highest growth opportunities.

Using a panel data approach, they find that the pecking order theory provides the prevailing reasoning behind the lower leverage used by tech firms throughout their life cycle. Information asymmetry turns growth opportunities and intangibles into the most differential drivers. In addition, differences in tech firms' strategic behaviour induce changes in their capital structure by stage, primarily during maturity, as well as during introduction and decline. These differences are closely related to the differential effect of the age variable, a sign of success, considering not only the rapid development but also the rapid decline of tech firms, both of which are linked to changes in growth rates. They find that growth opportunities are key factors for discriminating between groups of high-tech firms when studying their capital structure. Their results have relevant implications for researchers both in the selection of the sample to be analysed and the interpretation of results, depending on the distribution of the sample over the life cycle stages. Due to the evolution of the capital

structure patterns along the life cycle, a sample consisting of firms in different stages of their life cycles would produce some non-significant or spurious results, whereas different samples could produce opposite results, depending on the life cycle stage in which most firms are included. This emphasises why studying determinants of capital structure over a firm's life stages is significant.

This article differs from existing literature in several ways. First, it explains the capital structure through the firm's life cycle stages and observes the differences by stage, contributing to both research streams, i.e., capital structure and business life cycle, as this is the first time that Dickinson's (2011) approach is used in a capital structure article. Second, their work contributes to a small group of works comparing tech firms to non-tech firms concerning the capital structure. Unlike the two most comparable studies (Hogan & Hutson, 2005; Hyytinen & Pajarinen, 2005), which used a one-country small and medium-sized enterprise (SME) sample, they studied an international sample of listed companies. Moreover, the two works referenced adopt a static standpoint without considering life cycle stages. Third, they use the differences in growth opportunities to further discriminate between high-tech firms to analyse their capital structures.

A relevant study in the examination of common stages in the early literature is that by Miller and Friesen (1984). They found convergence in five life stages: birth, growth, maturity, revival and decline. The proxies used for the life cycle stages have changed quite a bit over time: investment (Wernerfelt, 1985), production behaviour (Wernerfelt, 1985), and learning and experience (Spence, 1981). In the context of financing decisions, Hirsch and Walz (2011) analysed SME's cycle considering two stages: start-up (defined as the period of product development) and expansion (defined as the period of market entry for the firm's product). Additionally, Pfaffermayr, Stöckl, and Winner (2013) used age to analyse the evolution of taxation and the capital structure. Our reference in the distinction of life stages, Dickinson (2011), uses the signs of operating, investing and financing cash flows. By considering these three aspects of the business jointly, this method of classifying the life cycle stages overcomes the partiality of using just one discriminant variable, which is commonplace in the literature.

The specific characteristics of the businesses in the tech sector produce pronounced differences in financial behaviour, mainly during maturity, followed by introduction and decline. Consistent with their first set of hypotheses, growth opportunities, amortisation of intangibles and age are the primary determinant factors for these stages, complemented by non-debt tax shields. The negative (or lower) effect of intangible assets throughout the life cycle, except at the introduction stage, and the positive effect of growth opportunities during growth and maturity phases, confirm the link between information asymmetry and risk in inducing tech firms to use more equity instead of debt. The positive coefficients of non-tech firms support the pecking order theory, and the differential coefficients of tech firms (lower or negative) for intangibles and growth opportunities are also addressed by the pecking order theory.

During the introduction and decline, age acts as a more potent and positive inducer of debt, pointing to the trade-off reasoning of reputation as prevalent. In these stages, the generation of internal funds is remarkably insufficient to cover the funding needs due to the negative operating cash flows. By contrast, age becomes a negative inducer of debt during maturity, in line with the pecking order reasoning, because, unlike non-tech firms, the sharp reduction of growth opportunities in tech firms takes place as firms maintain the positive generation of operating cash flows.

The abovementioned results on the effect of leverage drivers by life cycle stages confirm the role of information asymmetries in differentiating tech firms' capital structure patterns from those of non-tech firms. Furthermore, their results using the Frank and Goyal (2003) test of the pecking order theory confirm the significantly lower use of debt by tech firms than by non-tech firms across all the life cycle stages. In addition, the coefficients obtained for maturity support the previous results on the differential strategic role of this stage for tech firms, closer to that of shake-out for non-tech firms. Finally, Castro et. al. (2015) found that growth opportunities are key features for further distinguishing high-tech firms into smaller homogeneous groups to explain their capital structure better. Consequently, the pecking order is even more prevalent for high-tech firms with low growth opportunities (LGOs as they codify) across all the stages, and the study found that LGOs were closer to non-tech firms, in line with their second hypothesis.

The study contributes to the empirical literature on the capital structure in two ways. First, it explains the capital structure of tech versus non-tech firms along the life cycle stages. By doing so, our work confirms that Dickinson's (2011) model provides the research community with a new proxy for the life cycle, allowing us to apply capital structure models within a dynamic framework. This framework enables more detailed analyses of either general or specific sector samples. Our results highlight the importance of selecting homogeneous groups concerning life cycle stage to form the sample under analysis, thereby explaining capital structure theories more effectively. Thus, by using by-stage samples, the offsetting effect of some drivers that evolve along the life cycle is avoided, and some mixed effects found in the literature can be disentangled. Second, they demonstrate the potential for growth opportunities by identifying smaller groups of high-tech firms with distinct capital structures.

Castro et. al.'s (2016) paper analyses differences in target leverage and speed of adjustment across three life cycle stages of European listed firms: introduction, growth and maturity. The study determines that profitability and tangibility are the most stable determinants, whereas growth opportunities and size exhibit changing effects across stages. The speed of adjustment does not increase as the firms evolve, as firms in the introduction stage can adjust the fastest. Firms in the changing stage adjust their leverage at a slower pace, and their target is more influenced by profitability, particularly when transitioning from growth to maturity. Finally, the study confirms the existence of long-term debt targets by providing evidence that the next-year target is a relevant factor to explain current debt when firms change from one stage to another.

They used firm balance sheets, income statements, cash flow statements, and annual data from the Worldscope database. The cash flow statement is generally unavailable before 1989; therefore, the sample covers 1990-2012. The panel data contain all quoted firms from Austria, Belgium, Denmark, Finland, France, Germany, Greece, Italy, the Netherlands, Norway, Portugal, Spain, Sweden and the UK (11,553 firm-year observations). The sample comprises the leading European countries in terms of the number of listed firms.

Castro et al. (2016) categorise all firms into the introduction, growth, or maturity stages based on the classification criteria outlined in the "Measure of life cycle" section, primarily drawing from Dickinson's (2011) model. For their second and fourth hypotheses, they work with five sub-samples: firms that transition from introduction to growth (453 firm-year observations), firms that move from growth to maturity (1363 firm-year observations), and firms that remain in the same stage (837 firm-year observations in introduction, 2700 in growth, and 1683 in maturity). They exclude financial and regulated firms (SIC codes 6000-6999 and 4900-4999) from the sample. Moreover, they winsorise all variables at the 3% level to avoid the influence of outliers.

3.2.4. Literature providing Indian evidence

We do not come across studies focused on Indian firms covering aspects of financing over corporate life stages. This study, in that sense, acquires greater importance because of this important gap in the literature. Although we do not assume that we would find something uniquely different when we study determinants of capital structure over corporate life stages, studies focusing on firms in other countries have not covered all the stages of a firm's life. Therefore, we expect to find something that has not been previously identified, particularly in financing behaviour at startup stages and stages of decline and decay.

3.3. Chapter summary

The confirmation of Dickinson's model as a good proxy for assigning firms into the proper life cycle stage is also relevant for both researchers and practitioners. As in the case of capital structure, a good number of business decisions should be taken consciously of the changing (internal and external) conditions across the stages. For example, financial managers could use this model to assess their firms' life stage and compare their capital structure with that of the industry (or specific firms) at the same stage. Considering our second contribution, which confirms the long-term planning of relevant capital structure changes, financial managers can compare their firm's current capital structure with that of firms in the next stage of evolution they plan to achieve in the future. Thus, managers could anticipate some limitations and/or advantages arising

when adjusting the capital structure to face the challenges, meaning investment/disinvestment in the new stage

Research gaps

As far as this part of the world is concerned, by which we mean the developing economies of the world, including India, this study is an initiation of research in this domain. Of course, it is a humble effort. We find no study in India, to the best of our knowledge, focusing on determinants of capital structure over corporate life stages. Similarly, there is no work on determinants of changes in capital structure over corporate life stages. The aspects could relate to determinants or coefficients of determinants. It may also focus on which determinants are more relevant during which stages of corporate life. Moreover, it may also relate to how determinants may be different if our focus shifts from capital structure to changes in capital structure. Also, because financial leverage decisions are sticky, are these decisions equally sticky over different life stages of the corporates.

Conclusions

Our endeavour in this chapter has been to look at the relevant literature to identify the research gaps in the domain. We have identified potential areas of study for the next couple of chapters, including the sample to be used, the methodology to be applied, and aspects to be examined regarding the determinants of capital structure across corporate life stages.

Chapter 4
Research Design

CHAPTER 4

RESEARCH DESIGN

In this chapter, we turn to research design. Building on the insights drawn from the review of literature, this chapter outlines the research methodology adopted to investigate the core questions of the study. While the previous chapter examined existing theories, empirical findings, and conceptual perspectives relevant to the research problem, the current chapter transitions from what is known or understood to how this study aims to generate new understanding/findings. It details the research design, approach, data collection methods, sampling techniques, and describes the analytical tools to be employed. By grounding methodological choices in the gaps and patterns identified in the literature, this chapter ensures coherence between theoretical foundations and the empirical strategy used to address the study's objectives: determinants of capital structure with corporate life stages as a variable of interest.

4.1. Introduction

Research design is concerned with identifying research objectives, developing research hypotheses, listing variables to be used, selecting sample data, and applying appropriate methodology. The present study is based on secondary data, derived from financial statements of the companies listed on the BSE (Bombay Stock Exchange), as compiled and made available as a database service by CMIE (Centre for Monitoring of Indian Economy).

Having studied some preliminary, conceptual and empirical aspects related to the work, a research plan was developed to collect data relevant to the research questions, which were framed in the process of developing the research plan. Then those questions were transformed into researchable hypotheses. These hypotheses were then tested using sample data. Statistics, Econometrics, Computing tools, and statistical software (like STATA) have been deployed in the process.

The amount of information and data that is becoming available is growing exponentially. As the amount of available data grows, so does the need to understand

it. Statistics is the science of learning from data. Econometrics has been employed to examine economic relationships in the data, informed by a priori understanding derived from a literature review and analysis of phenomena related to corporate finance, specifically the capital structure of India's top non-financial corporate entities. Broadly, there are three main reasons why statistical literacy is essential in data science. First, it provides the skills to assess whether the data are sufficient to answer the questions. Second, it establishes a rigorous framework for quantifying uncertainty. Moreover, finally, it provides techniques for effectively communicating the findings of your analyses.

4.2. Research objectives

The research objective is to determine whether business firms differ in various life stages regarding financial leverage. Moreover, if the answer is affirmative, what explanations or determinants are identified or identifiable to such changes? So in that sequence, the study not only adopt a mechanism to assign different life stage value at different point of time according to aspects of business manifestations, it also looks at the determinants of financial leverage as explained by the stage in which the firm is at a particular point of time, but also use other variables as control variables to arrive at appropriate values of coefficients.

Across different corporate life stages, is financial leverage significantly different? The study considers both debt capital and equity capital, including reserves and funds, as total liabilities, and calculates the ratio of long-term debt to this total. Alternatively, one may take a ratio of long-term debt with equity capital and reserves and funds or long-term debt with long-term liabilities. We may coin these terms as LEVERAGE, LeverageLT, debt-equity ratio or debt to long-term capital ratio. Also, based on the cash flow patterns of cash flows from operating activities, investment activities and financing activities, following Dickinson's (2011) model, eight corporate life stages have been formulated.

4.2.1. Research questions

The research questions that may be relevant to investigate could be as follows:

To examine the relationship between corporate life stages and financial leverage.

To empirically find out the significance of corporate life stages as an explanatory variable.

The purpose of this study is to investigate the effects of the global financial crisis of 2008, the new bankruptcy regime implemented in 2016, and the global crisis caused by COVID-19, which affected the global economy from the beginning of 2020, using these as dummy variables on firms' capital structures.

To identify distinct determinants of financial leverage at each stage of corporate life.

4.2.2. Hypothesis development

There are three different perspectives when we develop our hypotheses: One, how the overall model looks. What are the variables that are important when we look at the effect of corporate life stages over leverage, whether this be lag of dependent variables, or these are control variables, exogenous variables, instrument variables, or the variable of interest, that is, the corporate life stage? Two, each of the corporate life stages may have some specific variables that are important when explaining the relationship between leverage and the corporate life stage. Accordingly, the plan of the study proceeds: Chapter 5 focuses on the overall results, and Chapters 6 to 8 focus on different stages from startup, growth, and maturity to decline and decay. The work concludes with a chapter summarising the study's findings, recommendations, and limitations.

So, each relevant hypothesis is elaborated on as the study proceeds in these chapters. Chapter 5, which is immediately the next chapter, studies whether long-term financial leverage is different across corporate life stages, and if so, what are the explanatory variables?

4.2.3. Research hypotheses

Our primary research hypothesis is that financial leverage varies across different corporate life stages. Additionally, there may be different determinants at various stages of corporate life. However, debt-financing decisions are sticky and not elastic enough to change very frequently. Therefore, it becomes more important to examine the determinants of capital structure, as well as those of changes in capital structure. These aspects of the study are examined in Chapter 5. The following are our other research hypotheses in chapters 6, 7 and 8:

Research Hypothesis # 6.1: Profitability (**prof**), Asset size (**logsize and size**), dividend payout (**dvnd**), tax shield (**taxShield**), global financial crisis (**GFC**), bankruptcy regime (**ibc2016**) and the global health crisis (COVID-19) (**dcovid20less**) market rate of interest (**intRate**), and market rate of return (**returnIndexClosing**) are determinants of capital structure at the startup stage.

Research Hypothesis # 6.2: Profitability (**prof**), Asset size (**logsize and size**), dividend payout (**dvnd**), tax shield (**taxShield**), global financial crisis (**GFC**), bankruptcy regime (**ibc2016**) and the global health crisis (COVID-19) (**dcovid20less**) market rate of interest (**intRate**), and market rate of return (**returnIndexClosing**) are determinants of changes capital structure at the startup stage.

Research Hypothesis # 7.1: Profitability (**prof**), Asset size (**logsize and size**), dividend payout (**dvnd**), tax shield (**taxShield**), global financial crisis (**GFC**), bankruptcy regime (**ibc2016**) and the global health crisis (COVID-19) (**dcovid20less**) market rate of interest (**intRate**), and market rate of return (**returnIndexClosing**) are determinants of capital structure at the growth stage.

Research Hypothesis # 7.2: Profitability (**prof**), Asset size (**logsize and size**), dividend payout (**dvnd**), tax shield (**taxShield**), global financial crisis (**GFC**), bankruptcy regime (**ibc2016**) and the global health crisis (COVID-19) (**dcovid20less**) market rate of interest (**intRate**), and market rate of return (**returnIndexClosing**) are determinants of capital structure at the maturity stage.

Research Hypothesis # 7.3: Profitability (**prof**), Asset size (**logsize and size**), dividend payout (**dvnd**), tax shield (**taxShield**), global financial crisis (**GFC**), bankruptcy regime (**ibc2016**) and the global health crisis (COVID-19) (**dcovid20less**) market rate of interest (**intRate**), and market rate of return (**returnIndexClosing**) are determinants of changes in capital structure at the growth stage.

Research Hypothesis # 7.4: Profitability (**prof**), Asset size (**logsize and size**), dividend payout (**dvnd**), tax shield (**taxShield**), global financial crisis (**GFC**), bankruptcy regime (**ibc2016**) and the global health crisis (COVID-19) (**dcovid20less**) market rate of interest (**intRate**), and market rate of return (**returnIndexClosing**) are determinants of changes capital structure at the maturity stage.

Research Hypothesis # 8.1: Profitability (**prof**), Asset size (**logsize and size**), dividend payout (**dvnd**), tax shield (**taxShield**), global financial crisis (**GFC**), bankruptcy regime (**ibc2016**) and the global health crisis (COVID-19) (**dcovid20less**) market rate of interest (**intRate**), and market rate of return (**returnIndexClosing**) are determinants of capital structure at the decline stage.

Research Hypothesis # 8.2: Profitability (**prof**), Asset size (**logsize and size**), dividend payout (**dvnd**), tax shield (**taxShield**), global financial crisis (**GFC**), bankruptcy regime (**ibc2016**) and the global health crisis (COVID-19) (**dcovid20less**) market rate of interest (**intRate**), and market rate of return (**returnIndexClosing**) are determinants of capital structure at the decay stage.

Research Hypothesis # 8.3: Profitability (**prof**), Asset size (**logsize and size**), dividend payout (**dvnd**), tax shield (**taxShield**), global financial crisis (**GFC**), bankruptcy regime (**ibc2016**) and the global health crisis (COVID-19) (**dcovid20less**) market rate of interest (**intRate**), and market rate of return (**returnIndexClosing**) are determinants of changes in capital structure at the decline stage.

Research Hypothesis # 8.4: Profitability (**prof**), Asset size (**logsize and size**), dividend payout (**dvnd**), tax shield (**taxShield**), global financial crisis (**GFC**), bankruptcy regime (**ibc2016**) and the global health crisis (COVID-19) (**dcovid20less**) market rate of interest (**intRate**), and market rate of return (**returnIndexClosing**) are determinants of changes capital structure at the decay stage.

4.3. Variables to be used in the study

Based on the theoretical perspective and review of literature in chapters two and three, the following variables have been identified and used to examine the relationship between leverage and corporate life stages:

Financial leverage, also known as leverage, indicates the ratio of long-term debt to the business's long-term capital, which is calculated by adding debt and equity.

Profitability of the business, which is profit before depreciation, interest, tax and amortization as a ratio of total assets.

The size of assets, whether tangible or intangible, is a measure of the strength of the company.

The size of tangible assets is a key measure of a company's strength.

The dividend payment ratio has an impact on the internal financial resources available for investment in the business of a company

Promoters' shareholdings pledged to mobilise financial resources

Tax Shield, that is, the amount of interest tax shield

Macro-economic variables, like stock market index return, indicating the market rate of return, and the **rate of interest**.

Some of the variables represent the impact of the global financial crisis, the insolvency and bankruptcy code, and the Covid-19 global catastrophe caused by the coronavirus SARS-CoV-2. In January 2020, the disease spread worldwide, resulting in the COVID-19 pandemic.

The following table presents the important variables used in our study, their measurements, and the assigned symbols.

Table 4.1: The list of variables, their measurement and symbols assigned

S. No.	Variable	Measurement	Symbol
1	financial leverage	long-term debt to total of debt and equity, including reserves and funds	leverage
2	profitability	Profit before Depreciation, Interest, Tax and Amortisation/Total Assets (PBDITA/TA)	prof
3	Size of assets	Total assets	size
4	Tangible assets	Net Fixed Assets/Total Assets	tang
5	Log of Size of Assets	Natural log of size of assets; log (Total Assets)	logsize
6	Dividend payout	(Equity Dividend Income Distributed/Profit after tax) *100	dvnd
7	Interest Amount	PBIT-PBT	Intamt
8	Tax Shield	Intamt*0.34	taxShield
9	corporate life stage	based on the pattern of cash flows	corplifestage
10	rate of interest	It is a macro variable taken as a 10-year RBI bond yield	intRate
11	Index of adjusted closing price of BSE Sensex	Calculated from the Adjusted Close Price of the Sensex at the end of the financial year	inturnIndexClosing
12	global financial crisis	It is a dummy variable with a cutoff point at the year 2008	GFC
13	bankruptcy regime	It is a dummy variable with a cutoff point at the year 2016	ibc2016
14	Covid'19	It is a dummy variable with a period of 2021 to 2023, impacted adversely by Covid'19	dcovid20less
15	Promoters' shareholdings	At the end of the year, promoters' shareholdings as a percentage of total paid-up share capital	pmShare

4.4. Data and sample selection

Secondary data has been taken from the financial statements of companies in India. These are non-financial firms from the top 500 listed on the BSE. The sample was drawn from the database service 'ProwessOnWeb', provided by CMIE (earlier known as Centre for Monitoring of Indian Economy). This private service has arrangements with the Ministry of Corporate Affairs, Government of India.

We studied the data for 24 years for companies listed on financial markets like the Bombay Stock Exchange and traded regularly. These 24 years were from FY2001 to FY2024. There was data for 8677 firm years. To extract data, we used Prowess on the Web database service provided by the Centre for Monitoring of Indian Economy (CMIE).

4.5. Methodology for data analysis

4.5.1. Univariate analysis

Linear regression is a statistical method for analysing the relationship between a dependent variable and one or more independent variables.

Univariate linear regression analyses the relationship between a single independent variable (X) and a dependent variable (Y), aiming to find a line that best fits the data points, represented by the equation $Y = \beta_0 + \beta_1 X$. Most economic relationships cannot be fully explained with just one explanatory variable.

Regress performs ordinary least-squares linear regression. Regress can also perform weighted estimation, compute robust and cluster-robust standard errors, and adjust results for complex survey designs.

4.5.2. Multivariate regression analysis

Multiple linear regression is a statistical technique that uses multiple independent variables to predict the outcome of a dependent variable, extending simple linear regression to account for multiple predictors.

Multivariate linear regression, an extension of multiple linear regression, models the relationship between multiple dependent (response) variables and one or more independent (predictor) variables, aiming to predict or explain the behaviour of several outcomes simultaneously.

The current work employs multiple linear regression techniques on panel data, which comprises cross-sections of units across time.

4.5.3. Panel data

The types of data generally available for empirical analysis are time series, cross-section, and panel. In time series data, we observe the values of one or more variables over a period (e.g., GDP for several quarters or years). In cross-sectional data, values of one or more variables are collected simultaneously for several sample units, or subjects (e.g., crime rates for states and union territories in India for a given year).

The same cross-sectional unit (say a family, a firm, or a state) is surveyed over time in panel data. In short, panel data have space as well as time dimensions.

Panel data (also known as longitudinal data or, in some disciplines, as cross-sectional time series when there is an explicit time component). Panel datasets have the form $X[it]$, where $x[it]$ is a vector of observations for unit i and time t . The `xt` command indicates x for cross-section and t for explicit time component. The study uses a panel of non-financial firms out of the top 500 firms listed on the BSE.

A panel of data could be a balanced panel or an unbalanced panel. In the panel data literature, one will also come across the terms short panel and long panel. In a short panel, the number of cross-sectional subjects, N , is greater than the number of periods, T . In a long panel, it is T that is greater than N .

What, then, are the options? There are four possibilities:

1. **Pooled OLS model.** If there is data for 90 airlines, it would simply pool all 90 observations and estimate a “grand” regression, neglecting the cross-section and time series nature of the data.
2. **The fixed effects least squares dummy variable (LSDV) model.** Here, if there are 90 airlines, we pool all 90 observations, but allow each cross-section unit (e.g., airline in this example) to have its own (intercept) dummy variable.
3. **The fixed effects within-group model.** Here, one can also pool all 90 observations. However, each airline would express each variable as a deviation from its mean value and then estimate an OLS regression on such mean-corrected or “demeaned” values.
4. **The random effects model (REM).** Unlike the LSDV model, in which we allow each airline to have its own (fixed) intercept value, one assumes that the intercept values are a random drawing from a much bigger population of airlines.

4.6. Methods of estimation for panel data

There are several ways one may estimate the relationship. However, before estimating the relationship, one must first assume the nature of the relationship, generate estimates, and then test the validity of both the relationship and its estimates.

4.6.1. Pooled OLS estimation

Why Do the OLS Need Assumptions?

We assume that the equation to be estimated is valid for each observation. Therefore, there are T lines, one for each observation:

$$y_1 = \alpha + \beta x_1 + u_1$$

$$y_2 = \alpha + \beta x_2 + u_2$$

.....

$$y_t = \alpha + \beta x_t + u_t$$

.....

$$y_T = \alpha + \beta x_T + u_T$$

The above is a system of equations with T lines. The question is how many unknowns the system has. The system has Ty 's and Tx 's. These are the things we know. The unknowns are the α , β and T error terms. There are also *unknown T by T covariances*.

This means that we have more unknowns than lines. In other words, this system cannot be solved. We therefore either fix the α and β , and calculate the error terms, or we add assumptions to the model so that this system can be solved. However, notice that we are not interested in the error terms in the end. For this reason, we add assumptions to the model. These assumptions are important, and we will need to discuss them in detail because if any of the assumptions do not hold, then the solution that we find is not going to be appropriate.

4.6.2. What are our assumptions?

In the example, it is important to notice that we have made the following assumptions when solving our model. There will be more assumptions to add to the model when we have more than one regressor.

Assumption 1: The Model is Linear

The assumption that the model is linear allows us to have a simplified problem to solve. Indeed, if the model is linear, then it will be easy to minimise the objective function that we will create.

Assumption 2: Expected Value of the Error Term is Zero

This second assumption is, in a way, more important. The assumption is that the expected value of the error term is equal to zero and errors are not correlated with the regressor. This means that we are assuming we do not commit any systematic errors, and that errors cannot be predicted using the current set of information provided. In our opinion, this is the basis of the rationality of the approach.

Assumption 3: The Variance of the Error Term is Finite and Constant

The variance of the error term is finite and constant, and the errors are not correlated with each other. Errors are identically and independently distributed. This assumption is important as the variance of the estimator that we will be recovering depends crucially on this assumption.

Assumption 4: The Regressors are non-stochastic

Assuming that regressors are non-stochastic means that regressors are fixed in repeated samples. This stems from the idea that our model is derived from an experimental approach, where the regressors are under the researcher's control. This is far from the case in econometrics, and indeed, we will have to leave this assumption for later.

Assumption 5: The Error Term is Normally Distributed

In a way, we have assumed that the mean of the error term is equal to 0 and that the variance is constant. We now need a shape of the distribution of the error term, and the easiest way to do this is to assume that the error is typical. We do not need this assumption for recovering the parameters we are interested in; however, we do need this assumption for hypothesis testing later.

We will revisit these assumptions in mathematical form; however, this provides a fair introduction to understanding why assumptions are necessary and which ones are required.

Interpreting the parameters

The linear regression model we are building is probably the most popular tool for estimating the impact of a set of variables, X , on the dependent variable, y . The interpretation of the regression coefficients of even the simplest model is not immediate. It can lead to some difficulties, for several reasons, including the nature of both the dependent and independent variables (whether continuous or categorical), their unit of measure (whether taken as logarithms).

Therefore, let us interpret the coefficients of both types of variables, the intercept, one continuous variable and one categorical variable. Although the example we develop here is taken from a linear regression model, the approach we discuss applies to interpreting coefficients from a regression model, even without interaction terms, that is, the multiplication of variables.

Assume we have a linear regression model with two independent variables, also known as regressors, that we want to express using the following regression equation:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + u$$

We have data Y, X_1, X_2 for the observations $i=1, 2, \dots, n$, and we estimate $\beta_0, \beta_1, \beta_2$.

In this model:

Y is the dependent or response variable

X_1 is the first regressor variable

X_2 is the second regressor variable

u is the error term, which is unmeasured.

The parameters in the model are:

β_0 is the intercept

β_1 is the first regression coefficient

β_2 is the second regression coefficient.

One example would be a model where the number of people using the bus (Y) depends on the cost of the ticket (X_1) and whether the city is big (X_2). The number of people using the bus is measured in thousands, the ticket is measured in pounds, and $\text{city} = 0$ if the city is small, and $\text{city} = 1$ if the city is big.

Let us assume that the regression equation was estimated as follows:

$$\hat{y} = 4750.18 - 258.18X_1 + 846.82X_2$$

4.6.3. Interpreting the intercept

The intercept, β_0 , is interpreted as the value you predict for Y if all regressors, in our case X_1 and X_2 , are equal to zero. We would expect, on average, 4,750.18 people to use the bus in small cities, assuming they can use it for free. However, this is only a meaningful interpretation if it is reasonable to assume that both X_1 and X_2 are equal to zero, and if the data set included values for X_1 and X_2 that are equal to or near zero. If neither of these conditions are true, then β_0 has no meaningful interpretation; it links the regression line correctly. In our case, it is easy to see that X_2 sometimes is 0, but if X_1 , the ticket cost, never comes close to 0, then our intercept has no real interpretation.

4.6.4. Interpreting coefficients of continuous independent variables

X_1 is a continuous variable, meaning that β_1 can be taken as the difference in the predicted value of Y for each one-unit difference in X_1 , other things being equal, which in our case is if X_2 remains constant. In other words, this means that if X_1 differed by one unit and X_2 did not, Y would differ on average by β_1 units. In our example, cities where bus tickets are rupee 1 more expensive would have about 258 fewer people on average taking the bus, if the size of the city is similar.

4.6.5. Interpreting coefficients of categorical variables

Similarly, β_2 is interpreted as the difference in the predicted value of y for each one-unit difference in X_2 if X_1 remains constant. However, because X_2 is a categorical variable coded as 0 or 1, a one-unit difference represents a switch between categories. Therefore, β_2 is then the average difference in y between the category for which $X_2=0$, which should be regarded as the reference group, and the category for which $X_2=1$, which is the comparison group. Therefore, compared to small cities, we expect that 5,597 ($4,750.18 + 1,846.82$) people will be taking the bus if it is free in big cities. It is convenient to use the average X_1 to make this comparison.

4.6.6. Interpreting coefficients when predictor variables are correlated

It is also important to remember that the other variables influence each coefficient in a regression model. Predictor variables are nearly always associated, meaning that two or more variables may explain some of the same variation in y . Therefore, each coefficient does not measure the total effect of its corresponding variable on y , as it would if it were the only variable in the model. Instead, each coefficient represents the additional effect of adding that variable to the model, if the effects of all other variables in the model are already accounted for. This means that each coefficient will change when other variables are added to or deleted from the model. We will study an example of this problem later in the course.

4.6.7. Interpreting the variance

The variance of the estimator is another parameter to interpret. The point is that any regression estimates, b , are built upon a specific sample. Therefore, if a different sample of data were chosen from within the population of observations, the data points would be different, and, in turn, this would lead to different values of the estimates. It is desirable to have a measure of how reasonable these estimates of β are.

What do we mean by how good?

We say how good in the sense of having a measure of the precision of the estimators of β , that is, of b . This is because it is helpful to know how confident we can be in the estimates and to know if and by how much these estimates are likely to vary if the sample changes, given the population. A good measure of the precision of the

estimates can be calculated using only the sample of data itself. This estimate is given by its standard error, which we know to be equal to:

$$SE(b) = \sqrt{2V((u'u)/(T-k)) * (X'X)^{-1}}$$

The standard errors are a function of the actual observations on the explanatory variable (X), the sample size (T), and another term ($s_2 = u'u/(T-k)$), that is, the estimate of the variance of the disturbance term. From the formula, we can see that the larger the sample size (T), the smaller the coefficient standard errors. This is clear, as the matrix X is a $T \times k$ matrix, because we assume that every observation represents a piece of helpful information that can be used to help determine the coefficient estimates. Therefore, the larger the sample size, the more information used in parameter estimation, and hence the greater the confidence placed in those estimates. Also, again from the formula, we can say that the larger the error variance ($u'u/(T-k)$), the more dispersed the residuals, and the higher the uncertainty in the model. If s_2 is larger, the data points are collectively far from the line.

The sum of the squares of the X 's about their mean contributes to determining the coefficients. More specifically, the larger the sum of squares, the smaller the coefficient variances. This means that if the X 's are dispersed around their mean, they lead to a more precise estimate. Notice that the sum of squared X 's only impacts the intercept. The standard errors give only a general indication of the likely accuracy of the regression parameters. They do not show how accurate a particular set of coefficient estimates is. If the standard errors are minor, it indicates that the coefficients are likely to be precise on average, rather than for this sample. Thus, standard errors give a measure of the degree of uncertainty in the estimated values for the coefficients.

4.6.8. Test for the violations

Recall that we made six assumptions relating to the classical linear regression model: linearity (A1), full rank (A2), regression model (A3), spherical errors (A4), non-stochastic regressors (A5) and standard errors (A6).

Five of them (A1 to A5) are required to show that the estimation technique, ordinary

least squares (OLS), had several desirable properties, that is, it is unbiased and efficient, and to ensure that hypothesis tests on the coefficient estimates could be conducted validly. When we incur a violation of one of the above assumptions, it means that what we assume in our model does not hold in our data.

4.6.9. Process of working out with a violation

Step one

Firstly, we need to identify why the assumptions in our data do not hold; that is, we must determine whether and what type of violation is present in our data.

Step two

Secondly, we need to focus on the consequences of the violation on our estimation. We can have three problems: the coefficient estimates are wrong, the associated standard errors are wrong, and the distribution that we use for testing hypotheses is inappropriate.

In all these cases, it will be impossible to make reliable inferences using our estimated parameter.

What if the coefficient estimates are wrong

The first case refers to the violation of assumption A3 (regression model). Recall that this assumption claims that $E(\varepsilon)=0$. The assumption implies that the X s are not correlated with the error term. We used this assumption to demonstrate that the parameter estimate obtained via OLS is, on average, equal to the actual parameter, indicating that the parameter is unbiased. We also know that unbiasedness is necessary to safely conduct the test hypothesis on our parameter, being the estimate for β , the numerator of the t-test formula. So, if the assumption A3 is violated, the parameter is not unbiased anymore, which means that any inference made using that parameter is wrong, simply because the value of the parameter is wrong. The most suitable correction for addressing the correlation between the X s and the error term is the Instrumental Variable estimator.

What if the associated standard errors are wrong

The second problem deals instead with assumption A4 (spherical errors), which assumes homoscedasticity and non-autocorrelation of the errors. We used this assumption to show that the OLS estimator is efficient; it has the least variance among the linear and unbiased estimators. Recall that the $\text{var}(\hat{\beta}) = \sigma^2(X'X)^{-1}$, so it is intuitive that, if the assumption A4 is violated, the main consequence will be on the efficiency of the OLS estimator. This, in turn, being the variance of the OLS estimator at the denominator of the t-test formula, invalidates the inference procedure.

The distribution used for testing hypotheses is inappropriate

The last problem refers to the assumption that the A6 error is not normally distributed, $\varepsilon \sim N(0, \sigma^2)$. If the error does not follow a normal distribution, the OLS estimator, which is proved to be a linear function of the error term, also does not follow a normal distribution. Therefore, the inference we make based on the t-test will be wrong.

Step three

The last step refers to finding the most suitable solution to the problem. At this stage, we can either eliminate the violation, if possible, or, if the violation cannot be eliminated, we can use alternative techniques which are still valid. The process of correcting for violations involves the following steps: Identify the technique needed for detecting violations; Find out the source of the violation, which is why we have a violation; understand the consequences of having a violation on our estimation; and identify the most appropriate technique to eliminate the source of the violation.

The procedure used to detect a violation resembles the procedure we apply for testing hypotheses on parameters, with the aim being different, however. The testing procedure applied to detect a violation is called diagnostic testing, which detects any misspecification in our model. Similarly, to any test hypothesis, diagnostic tests are based on the calculation of a test statistic. There are different approaches at an econometrician's disposal to build up the test statistic. Depending on the approach used, the test statistic will follow a particular distribution.

Econometric analysis of panel data

The term "panel data" refers to the pooling of observations on a cross-section of households, countries, firms, or other units over several periods. This can be achieved by surveying several such units and following them over time. These are known as micro-panels and are collected for a large number of N individuals (usually in the hundreds or thousands) over a short period T (varying from a minimum of two years to a maximum rarely exceeding 10 or 20 years). In contrast, macro-panels typically encompass a broader range of countries, such as the G7 countries, a larger set of 20 OECD or European Union countries, or the G20 countries, or a combination of developed and developing countries, which can be as extensive as 100 or 200. These are usually observed annually over 20 to 60 years. Micro- and macro-panels require different econometric care. For example, the asymptotics for micro-panels must be for large N and fixed T . In contrast, the asymptotics for macro-panels can be considered for large N and T . Additionally, with a long time series for macro-panels, one must address issues such as non-stationarity in the time series, including unit roots, structural breaks, and cointegration. In contrast, for micro-panels, one does not deal with non-stationarity issues, especially since T is short for each household or business firm. Also, in macro-panels, one must deal with cross-country dependence. These are not usually an issue in micro-panels, where households are randomly sampled and hence are unlikely to be correlated with, or these are business firms, each of which has unique firm-specific characteristics.

Data used

The Centre for Monitoring of Indian Economy (CMIE), previously known by its former name, was the primary source of our data. We used S&P BSE 500 data for 24 financial years, filtering it to obtain data for a set of 401 non-financial private sector firms.

Data has been elusive to the researcher, not because it is not there, but due to two-fold reasons. One reason is that the researcher was cautious as to its availability while reviewing the literature; the researcher updated it many times as the literature on the subject revealed the importance of different variables over time. However, also, because we updated it many times, it became somewhat unwieldy at times.

4.7. Models for determinants of capital structure

As stated earlier, we have used panel data for 401 non-financial firms using annual financial data from FY2001 to FY2024. We tested various models, including fixed-effects, random effects, and dynamic models, with the inclusion of lags of dependent variables. A brief description of various models, their comparison, and the model selection process may be in order here.

4.7.1. Panel data results

Panel data regression is a statistical technique used when analysing data that combines both cross-sectional and time-series elements. It is beneficial for studying changes over time while considering variations across different entities.

Panel data, also known as longitudinal data, consists of observations on the same entities (e.g., individuals, companies, countries) over multiple periods. This type of data allows researchers to track the dynamics of variables and relationships over time, while also accounting for individual differences or variations between entities.

Why use panel data regression?

- **Controlling for unobserved heterogeneity:** Panel data regression allows for the inclusion of unobserved, time-invariant characteristics of each entity (e.g., individual preferences, company culture). This can help mitigate bias that would occur if these unobserved factors were not accounted for in traditional regression models.
- **Examining the dynamics of relationships:** Panel data allows for the analysis of how variables evolve and how relationships between variables might change over time.
- **Addressing multicollinearity:** Panel data can help address multicollinearity, which occurs when independent variables are highly correlated, by analysing the relationships between variables at different points in time.

- **Improved efficiency and accuracy:** By utilising multiple observations over time, panel data regression can lead to more efficient and accurate estimates of the parameters of interest.

Types of panel data regression models

- **Pooled regression:** This model treats all observations as if they are from a single cross-section, ignoring the time-series dimension of the data. It is the simplest type of panel data regression if the relationship between variables remains consistent across all entities and periods.
- **Fixed effects regression:** This model accounts for the unobserved, time-invariant characteristics of each entity by including entity-specific intercepts in the regression. It enables the analysis of how variables evolve while controlling for entity-specific differences.
- **Random effects regression:** This model assumes that the entity-specific intercepts are randomly drawn from a population, and the model estimates the variance of these intercepts. It is more appropriate when the entity-specific effects are assumed to be random and not correlated with the regressors.

Key considerations

- **Data structure:** Panel data can be balanced (each entity has the same number of observations across time) or unbalanced (entities have different numbers of observations).
- **Model selection:** The choice of model depends on the research question, the nature of the data, and the assumptions made about the unobserved entity-specific effects.
- **Statistical tests:** Various statistical tests are available to help determine the appropriateness of a particular model, such as the Hausman test and the Chow test.

In summary, panel data regression is a powerful tool for analysing data that combines cross-sectional and time-series dimensions, allowing researchers to study changes over time while controlling for individual differences and unobserved heterogeneity.

4.7.2. Robustness

A robustness check is a validation technique used in statistical analysis and machine learning to ensure that findings are not overly reliant on specific assumptions or data inputs. It involves re-estimating models or applying alternative methods to see if the results remain consistent.

Here is a more detailed explanation:

Purpose of robustness checks

- **Verification of results:** Robustness checks help confirm that the main conclusions remain valid even when assumptions or data inputs are altered.
- **Identifying sensitivities:** They highlight how sensitive the conclusions are to different model parameters or choices.
- **Ensuring generalizability:** Robustness checks can help extend findings beyond the specific sample used in the initial analysis, enhancing the external validity of the research.

How Robustness Checks are Performed:

- **Alternative specifications:** re-estimate models using different variables, model structures, or statistical techniques.
- **Subsets of data:** analyse different subgroups of the data or exclude outliers to see how the results are affected.
- **Changes in assumptions:** Robustness checks can explore how the results would change if certain assumptions about the data or model were relaxed or violated.

Examples of robustness checks

- In economics, researchers might perform robustness checks by using different control groups or measures of the treatment variable.
- In machine learning, they might test the model's performance under adversarial attacks or with perturbed data.
- In statistical analysis, they might test for heteroscedasticity or other violations of assumptions.

In essence, a robustness check is a method for building confidence in a study's findings by demonstrating that the results are not artefacts of specific choices or assumptions.

4.7.3. Fixed effects model

Fixed effects regression is a statistical method used in panel data analysis to account for unobserved individual-specific characteristics that may be correlated with the independent variables, potentially leading to biased results in standard regression models. By including fixed effects, the model effectively controls for these unobserved, time-invariant factors, allowing for a more accurate estimation of the causal effect of the independent variables on the dependent variable.

Key aspects of fixed effects regression:

- **Panel data:** This method is designed explicitly for datasets that have repeated observations on the same individuals over time (panel data).
- **Unobserved individual-specific effects:** Fixed effects account for individual-specific characteristics that remain constant over time but may still influence the outcome variable.
- **Demeaning:** The method often involves demeaning the data (subtracting the individual's mean) to remove the effects of these time-invariant factors.
- **Bias reduction:** By controlling for unobserved effects, fixed effects regression helps to reduce bias in the estimation of the coefficients of the independent variables.
- **Fixed effects are estimated:** In fixed effects models, each group's mean is treated as a fixed parameter, rather than a random sample from a population, as in random effects models.
- **Within-entity variation:** Fixed effects regression primarily focuses on changes within each individual or entity over time, controlling for their stable characteristics.

- **Dummy variables:** Fixed effects can be thought of as including dummy variables for everyone or a group in the regression model.

When to use fixed effects regression:

- When dealing with panel data, one suspects that unobserved, time-invariant characteristics may be correlated with the independent variables.
- When the goal is to estimate the causal effect of the independent variables on the dependent variable, controlling for individual-specific effects.
- When the data contains repeated observations on the same individuals over time.

4.7.4. Random effects model

Random effects regression is a statistical method used when analysing clustered or panel data, where observations within groups are not independent. It assumes that there are random, unobserved effects specific to each group, which are incorporated into the model. This allows for the estimation of group-specific intercepts or slopes, acknowledging that observations within the same group may be correlated.

Here is a more detailed explanation:

- **Clustered or panel data:** Random effects models are beneficial when data are collected from multiple groups (e.g., students within schools, patients within hospitals, repeated observations of the same individual).
- **Unobserved group effects:** The model acknowledges that there might be unobserved factors specific to each group that influence the outcome.
- **Random effects:** These unobserved group effects are assumed to be random and uncorrelated with the independent variables in the model.

Benefits

- **Accounting for dependence:** Random effects models address the problem of within-group correlation, which is a common issue in clustered or panel data.
- **More accurate estimates:** By accounting for group-specific effects, the model can provide more accurate estimates of the overall effects of the independent variables.

- **Time-invariant variables:** Random effects models can handle time-invariant variables (e.g., province characteristics) that are absorbed by the intercept in fixed effects models.
- **Generalizability:** Random effects models allow for inferences to be generalised beyond the specific groups included in the study.

Examples

- **Meta-analysis:** Random effects models can be used in meta-analysis to account for heterogeneity among study results.
- **Panel data analysis:** Random effects models are used in panel data analysis to study the effect of individual-specific characteristics on the response variable.
- **Longitudinal studies:** They are also used in longitudinal studies to model correlation among repeated measures from the same subjects.

Key differences from fixed effects models

- **Fixed effects models** treat group effects as fixed parameters to be estimated, while random effects models treat them as random variables drawn from a distribution.
- **Fixed effects models** focus on within-group variation, while random effects models also consider between-group variation.
- **Fixed effects models** can be less efficient than random effects models if the random effects assumption holds

4.8. Model comparison and selection process

4.8.1. Model selection process

Having estimates of fixed effects and random effects contributes to the process of comparing and selecting an appropriate model. The Hausman test is one such tool.

4.8.2. The Hausman test

The Hausman test, also known as the Durbin-Wu-Hausman test or Hausman specification test, is a statistical hypothesis test in econometrics used to determine the

consistency and efficiency of estimators. It is often used in panel data analysis to determine whether to use fixed effects or random effects models. The test compares the estimates from two different estimators: one is assumed to be both consistent and efficient, while the other is consistent but potentially less efficient.

Here is a more detailed explanation:

Purpose:

- **Consistency vs. efficiency**

The Hausman test helps determine if the difference between the estimates from two estimators is statistically significant. If the difference is significant, it suggests that the more efficient estimator is preferred.

- **Endogeneity in panel data**

In panel data analysis, it is used to determine whether to use fixed effects or random effects models. The null hypothesis is that the random effects model is appropriate, while the alternative is that the fixed effects model is necessary.

- **Testing exogeneity**

The test also assesses whether the unobserved individual effect is correlated with the conditioning regressors. If the test rejects the exogeneity assumption, it suggests that a fixed effects model is more appropriate.

How it works

1. **Two estimators**

The test compares estimates from two estimators: one that is both consistent and efficient, and another that is consistent but potentially less efficient.

2. **Null hypothesis**

The null hypothesis assumes that the efficient estimator is the better choice and that the two estimators provide similar results.

3. Alternative hypothesis

The alternative hypothesis assumes that the less efficient estimator is necessary due to the correlation between the unobserved individual effect and the conditioning regressors.

4. Test statistic

The test statistic (Hausman statistic) is calculated based on the difference between the estimates and their covariance matrices.

5. P-value

The p-value is calculated based on the test statistic and the null hypothesis. If the p-value is below a certain significance level (e.g., 0.05), the null hypothesis is rejected, and the alternative hypothesis is supported.

4.8.3. Breusch pagan Lagrange multiplier test

In statistics, the **Breusch–Pagan test**, developed in 1979 by Trevor Breusch and Adrian Pagan, is used to test for heteroskedasticity in a linear regression model. It was independently suggested with some extension by R. Dennis Cook and Sanford Weisberg in 1983 (**Cook–Weisberg test**). Derived from the Lagrange multiplier test principle, it tests whether the variance of the errors from a regression is dependent on the values of the independent variables. In that case, heteroskedasticity is present.

4.9. Dynamic panel estimation (System GMM estimator)

The System GMM (System Generalised Method of Moments) estimator is a statistical method used to estimate dynamic panel data models, particularly those with endogenous regressors. It addresses endogeneity issues by using lagged levels and lagged differences of the dependent and independent variables as instruments. The system GMM estimator combines moment conditions from both the first-differenced model and the level model, offering potential advantages over the first-differenced GMM estimator.

Here is a more detailed explanation:

- **Addressing endogeneity**

In dynamic panel data models, the lagged dependent variable is often correlated with the error term, leading to biased and inconsistent estimates when standard methods are employed. System GMM tackles this by utilising lagged levels and differences of the endogenous variables as instruments.

- **Combining moment conditions**

The system GMM estimator combines moment conditions from two equations: the first-differenced model (which helps address individual-specific effects) and the level model (which uses lagged levels as instruments).

- **Benefits**

The system GMM estimator is known to have lower bias and higher efficiency than the first-differenced GMM estimator, especially when dealing with persistent data.

- **Two-step process**

The estimation process typically involves two steps:

1. **First-differencing:** The variables are first-differenced to eliminate individual-specific fixed effects.
2. **Instrumenting:** Lagged levels and lagged differences of the endogenous variables are used as instruments for the differenced equations.

- **Instruments in different equations**

The instruments used in the system GMM estimator can vary across equations. For instance, additional lags of the instruments are available in later periods and can be used.

In summary, the system GMM estimator is a powerful tool for analysing dynamic panel data models where endogeneity is a concern. It combines moment conditions from both the first-differenced and level models, potentially leading to more accurate and efficient estimates, especially when dealing with persistent data.

4.9.1. The test for overidentification

Sargan–Hansen test, or Sargan's test, is a statistical test used for testing over-identifying restrictions in a statistical model. It was proposed by John Denis Sargan in 1958, and he derived several variants in 1975. Lars Peter Hansen reworked the derivations and showed that they can be extended to general non-linear GMM in a time series context.

The Sargan test assumes that model parameters are identified via a priori restrictions on the coefficients and tests the validity of over-identifying restrictions. The test statistic can be computed from residuals from an instrumental variables regression by constructing a quadratic form based on the cross-product of the residuals and the exogenous variables. Under the null hypothesis that the over-identifying restrictions are valid, the statistic is asymptotically distributed as a chi-square variable with degrees of freedom ($L - K$, where L is the number of instruments and K is the number of endogenous variables).

4.9.2. The test for serial correlation

The Durbin-Watson test is the primary statistical test used to detect serial correlation in a dataset, specifically for first-order serial correlation (correlation between consecutive observations). This test examines the residuals from a regression model to see if there is a significant relationship between them and their lagged values.

4.10. Chapter summary

This chapter sets the tone for the study's conduct and conclusions. It delves into the data and methodology used to reach the results. It summarises important methods relevant to our research. In particular, the study will employ pooled OLS, fixed effects (FE), random effects (RE), and dynamic panel data analysis. We will also conduct post-estimation tests to evaluate the choice of estimation results and the validity of our findings.

Chapter 5
Capital Structure and Corporate Life
Stages
